

Measuring Database Unfairness Via Dependency Quantification Under Differential Privacy

Mariia Vologdin
Hebrew University
mariia.vologdin@mail.huji.ac.il

Yuchao Tao
Independent Researcher
harry.t.chao@gmail.com

Amir Gilad
Hebrew University
amirg@cs.huji.ac.il

ABSTRACT

Differential privacy (DP) has become the de facto standard for protecting sensitive data, providing strong guarantees that published statistics or models reveal limited information about any individual. However, privacy noise and restricted data access make it increasingly difficult to assess the fairness and reliability of private datasets. In this paper, we propose a formal framework for quantifying data unfairness under DP. We identify three core desiderata for unfairness measures based on previous work: positivity, monotonicity, and DP computability. We further instantiate them through three complementary measures: (1) a mutual information-based measure with a total variation distance proxy suitable for DP, (2) a data-repair-based measure approximated via a reduction to weighted MaxSAT, and (3) a top- k tuple contribution measure that isolates the most influential records in fairness violations. We design privacy-preserving algorithms and analyze their sensitivity, accuracy, and efficiency. Extensive experiments on multiple real-world datasets demonstrate that our proposed measures faithfully approximate their non-private counterparts, effectively quantify bias under privacy constraints, and provide insights for fair data management.

1 INTRODUCTION

Differential privacy (DP) [23, 26] has become the leading standard for safeguarding sensitive information in data analysis and machine learning (ML). It gives formal guarantees that published statistics or trained models reveal only limited information about any individual, even in the presence of auxiliary knowledge. DP has been widely adopted in practice by major organizations [21, 27, 76] and governmental agencies [2, 13, 31], and has been applied to a variety of tasks including private query answering [22, 38, 40, 51, 54, 63], synthetic data generation [6, 14, 32, 55, 56, 80, 87], and private machine learning [1, 44, 69]. Despite its broad success, DP introduces a fundamental challenge: while it ensures privacy, the injected noise and limited data access make it difficult to evaluate the *quality* and *fairness* of private data. Users and analysts must often rely on privacy-protected datasets to make decisions or train models, without being able to assess whether these datasets are accurate, representative, or equitable. Furthermore, learning from noisy or biased data can yield inaccurate and discriminatory predictions [73, 75].

Fairness and bias in data are central concerns in data management and ML, since biased data can produce unfair outcomes even when algorithms are well-designed. Extensive research has focused on defining algorithmic fairness, proposing formal notions such as Demographic Parity [11] and Conditional Statistical Parity [17], among many others [7, 24, 42, 48]. These notions constrain model behavior to ensure equitable treatment across protected groups. However, such definitions focus solely on the model, neglecting the data which itself may contain bias or unfairness. If the dataset

encodes biased relationships between protected and outcome attributes, an algorithm may reproduce or amplify those disparities.

In this work, we shift the focus from algorithmic fairness to the *fairness of data*. A key observation underlying our work is that many algorithmic fairness definitions can be expressed as forms of *conditional independence* [74]. For example, Demographic Parity requires independence between the sensitive attribute and the model outcome, while Conditional Statistical Parity introduces a conditioning set of admissible attributes. Using this formulation, we reinterpret fairness as a property of the *data distribution* itself rather than only of the learned classifier. This is motivated by a common assumption from a prior work [12, 29, 74]: if the labels in a dataset reflect the outcomes a reasonable classifier would produce, then dataset fairness becomes a reliable proxy for assessing the fairness of a classifier trained on it. Moreover, using the data in other manners, such as getting summarized statistics, may lead to biased conclusions. Consequently, testing whether a dataset satisfies such conditional independence statements provides a natural and principled way to assess its fairness. This connection enables the adaptation of well-established algorithmic fairness notions into *database-level fairness criteria*, which serve as the foundation for the framework we propose.

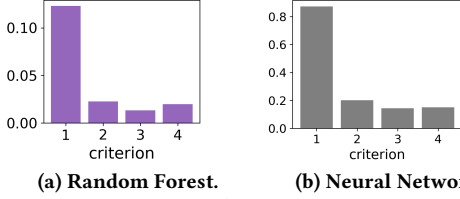
Our goal is to develop a principled, quantitative framework for measuring data unfairness that remains meaningful even when sufficient noise is added to them to satisfy DP. Prior research on measuring data quality has largely examined logical inconsistencies in data through integrity constraints, both without [8, 61, 70] and, recently, with privacy guarantees [66]. Yet, despite the growing literature on fairness in algorithms, no existing framework directly measures the unfairness of the private data itself, even though such measurements could give a valuable signal for data equity when the data cannot be fully observed.

Here, we introduce *differentially private unfairness measures* that, given a database D and a set of fairness criteria $\mathcal{F}$, compute a numerical score $\mathcal{U}(\mathcal{F}, D)$, quantifying how far the data deviates from satisfying those criteria. We formalize three key desiderata that any unfairness measure should satisfy, following principles devised for inconsistency measures [61]. First, the *Positivity* property [37, 70] ensures that the measure is non-negative and equals zero if and only if D satisfies all fairness criteria. Second, the *Monotonicity* property [70] states that expanding the set of fairness criteria ($\mathcal{F}_1 \subseteq \mathcal{F}_2$) cannot reduce the measured unfairness. Third, the measure must be efficiently and accurately computable under DP, maintaining interpretability and utility despite the added noise.

We present three complementary measures grounded in probabilistic dependence. The first, $\mathcal{U}_{\text{MI}}^{\text{TPD}}$, uses mutual information (MI) [18] to quantify dependence between sensitive and outcome attributes. Because MI is highly sensitive and thus unsuitable for



(a) Values of $\mathcal{U}_{MI}^{TVD}$. (b) Values of $\mathcal{U}_R^{SAT}$. (c) Values of $\mathcal{U}_{TC}$.
Figure 1: Values of the three unfairness measures (log scale) on Adult dataset for the four fairness criteria in Example 1.1.



(a) Random Forest. (b) Neural Network.
Figure 2: Fairness values (Demographic Parity and Conditional Statistical Parity) of the two privately-trained ML models on Adult dataset in Example 1.1.

DP [84], we develop a Total Variation Distance (TVD) proxy that closely approximates MI in both theory and practice. The second measure is based on data repair [16, 59, 74], quantifying the minimal number of tuple modifications required to make a dataset fair. Since this is computationally hard, we propose a proxy measure, $\mathcal{U}_R^{SAT}$, reducible to the weighted MaxSAT problem [82], following [74]. Finally, inspired by tuple contribution analysis [20, 57, 65, 83], we introduce a top- k contribution measure, $\mathcal{U}_{TC}$, that captures how much the most influential tuples contribute to fairness violations.

We prove that all three measures satisfy our desiderata, are close approximations to their originals, and exhibit low sensitivity relative to their range. We then design privacy-preserving algorithms for them that rely on the Laplace mechanism [25], and analyze their utility and complexity. Extensive experiments on five real-world datasets across diverse fairness settings confirm that our measures reliably capture comparative unfairness, remain faithful to their non-private baselines, and can be computed effectively under DP.

Example 1.1. Consider the Adult dataset¹ containing personal information of individuals in the US including whether their income is larger than 50K (we sample 10,000 tuples with 15 attributes). Figure 1 depicts the results of applying our three measures to four fairness criteria: (1) a person’s income should not depend on their years of education, (2) a person’s income should not depend on their sex, (3) a person’s income should not depend on their race, and (4) a person’s income should not depend on their sex, given that they work a certain number of hours per week (the criteria are phrased as independence statements in Table 2 in the sequel). We expect the first criterion to clearly be violated, as more education often leads to higher salaries, while the latter three criteria are natural and desired for most applications to avoid discriminatory conclusions and decision-making. Furthermore, it is known that Adult has disparity in income for different sexes and races [78, 81]. Each measure was evaluated using our DP algorithms with a privacy budget of $\epsilon = 1$, with the experiment repeated ten times and the

results averaged for each criterion. All the measures show a similar trend for criteria (1), (2), (3) and (4).

We additionally privately trained two models on the same sample using a budget of $\epsilon = 10$ and all available attributes to predict the outcome: (a) a DP version of RandomForest, which achieved 78.9% accuracy on average, and (b) a neural network with one hidden layer with 32 nodes and ReLU activation, trained with DP-SGD, which achieved 84.1% accuracy on average. Figure 2 shows the Demographic Parity gaps for criteria (1), (2) and (3), based on the classifier’s predictions: $|\max_{p \in P} Pr_D(O = 1 | P = p) - \min_{p \in P} Pr_D(O = 1 | P = p)|$ where $O = 1$ is a prediction of the income being bigger than 50K. Intuitively, this measures the disparity in positive outcomes for the most privileged group and the most discriminated one in a population consisting of more than two groups. The figure also shows the Conditional Statistical Parity gap for the conditional criterion (4), which is calculated as the expectation of the Demographic Parity over the conditioned attribute [71, 74]. We can see that the trend of our measures matches the trend of the ‘fairness measures’ commonly applied to ML models, indicating that the three measures are able to estimate the underlying unfairness based on the criteria, and that they might also be useful in predicting whether private data can be effectively used for various applications.

Contributions. The paper makes the following contributions:

- (1) We introduce a **formal framework for measuring data unfairness**, focusing on quantifying bias directly at the data level rather than at the algorithmic level, and defining general desiderata of *positivity*, *monotonicity*, and *DP computability*.
- (2) We propose **three concrete unfairness measures**, grounded respectively in (a) *mutual information* and its *total variation distance* proxy suitable for DP, $\mathcal{U}_{MI}^{TVD}$, (b) a *data repair* inspired measure, $\mathcal{U}_R^{SAT}$, approximated via a reduction to *weighted MaxSAT*, and (c) *tuple-level contribution* measure, $\mathcal{U}_{TC}$, identifying the top- k most influential records in fairness violations.
- (3) We design **DP algorithms** for computing the measures, and give formal analysis of their guarantees, error, and complexity.
- (4) We conduct an **extensive experimental evaluation** on five real-world datasets, demonstrating that our proposed measures faithfully approximate their non-private counterparts and effectively quantify data unfairness under privacy constraints.

2 PRELIMINARIES

We now give the necessary background for the paper, including basic database and probabilistic notions, fairness concepts, and essential definitions for our use of differential privacy.

2.1 Databases and Probabilities

We consider a single-relation schema $\mathcal{A} = (A_1, \dots, A_m)$, which is a vector of distinct attribute names A_i , each associated with a domain $Dom(A_i)$ of values. A database D over $\mathcal{A}$ is associated with a set $tids(D)$ of *tuple identifiers*, and it maps every identifier $i \in tids(D)$ to a tuple $D[i] = (a_1, \dots, a_m)$ in $A_1 \times \dots \times A_m$. We denote the size of the database by $|D| = n$. A tuple $t_i \in D$ has a specific value in each of its attributes, denoted by $t_i[A] = a$ where $a \in Dom(A)$. We consider bag semantics where t_i, t_j may share the same values

¹See the descriptions and links to the datasets in Section 5.1.

across all attributes except their identifiers. Sets of attributes are denoted by bold letters such as A and B .

The computation of empirical probabilistic quantities in databases is often practically performed by measuring database statistics [28, 41, 86]. Let D be a dataset of size n , the *empirical marginal* of a value $A = a$ is defined as follows: $Pr_D(A = a) = \frac{|\{t | t[A] = a\}|}{n}$. Similarly, the *conditional marginal* of $A = a$ conditioned on $C = c$ is

$$Pr_D(A = a | C = c) = \frac{|\{t | t[A] = a, t[C] = c\}|}{|\{t | t[C] = c\}|}$$

When a set of attributes P equals a set of corresponding values p , we abuse notation and write $P = p$, when clear from context.

Employing these notations, we review the definition of mutual information, which is commonly used to quantify dependencies between variables and will be the basis for one of our measures.

DEFINITION 2.1 (MUTUAL INFORMATION (MI) [18]). *The conditional mutual information between two attributes A and B conditioned on attribute C over their domain in a database D is defined as follows.*

$$MI_D(A, B | C) = \sum_{c \in \text{Dom}(C)} Pr_D(C = c) \sum_{a \in \text{Dom}(A)} \sum_{b \in \text{Dom}(B)} Pr_D(A = a, B = b | C = c) \log \left(\frac{Pr_D(A = a, B = b | C = c)}{Pr_D(A = a | C = c) Pr_D(B = b | C = c)} \right)$$

where C may be absent for the unconditional case.

We say that two attributes $A, B \in \mathcal{A}$ in a database D are independent if $Pr_D(A = a, B = b) = Pr_D(A = a) \cdot Pr_D(B = b)$ for all $a \in \text{Dom}(A)$ and $b \in \text{Dom}(B)$, or, equivalently, if $MI_D(A, B) = 0$, and we use the standard notation of independence $A \perp B$.

The notion of database probabilities can be used to express associational fairness definitions, as we discuss next.

2.2 Fairness as Independence Statements

Multiple fairness definitions that have been considered by previous work can be described as (conditional) independence statements. Demographic Parity [11] measures whether the favorable prediction of an outcome attribute $O \in \mathcal{A}$ (e.g., an income will be above a threshold) is affected by which the demographic group the person belongs to, as captured by a protected attribute $P \in \mathcal{A}$ (e.g., Race, Sex). This statement can also be expressed by $P \perp O$. Conditional Statistical Parity [17] further relaxes this definition by allowing certain admissible attributes $A \in \mathcal{A}$ (e.g., education level, work experience) to explain the unfairness, and can be written as $P \perp O | A$. Equality of Opportunity [42] defines fairness by stating that for each case of the ground truth (e.g., the loan is indeed approved), as captured by the response attribute $Y \in \mathcal{A}$, the probability of a favorable outcome should be equal across each demographic group, i.e., $P \perp O | Y$. Predictive Parity [15] measures whether predicted positives are equally accurate across demographic groups, or equivalently, whether the ground truth is independent of the protected attribute given the predicted outcome, i.e., $P \perp Y | O$.

We will simplify the discussion for our purposes and distinguish only between a protected attribute $P \in \mathcal{A}$, a response attribute $Y \in \mathcal{A}$, and an admissible attribute $A \in \mathcal{A}$. Thus, we can generally define a *database fairness criterion* as follows.

DEFINITION 2.2 (FAIRNESS CRITERION). *Given a schema $\mathcal{A}$ with a protected attribute $P \in \mathcal{A}$, response attribute $Y \in \mathcal{A}$, and admissible*

attribute $A \in \mathcal{A}$, a fairness criterion is an independence statement $P \perp Y | A$, where A may be absent.

A set of multiple fairness criteria will be denoted by $\mathcal{F}$.

Our task of defining unfairness measures for databases now becomes a general approach of quantifying the dependence of attributes in a database (we focus here on fairness notions that are consistent with Definition 2.2, but recognize that many fairness notions exist beyond it and plan on addressing them in future works). However, we must consider the fact that the databases we wish to analyze are protected with differential privacy.

2.3 Differential Privacy

We next give the necessary definitions for DP.

DEFINITION 2.3 (NEIGHBORING DATABASES). *Two databases D, D' with the same schema are called neighboring if they have the same size and differ in exactly one tuple. We denote this relationship by $D \sim D'$.*

We often use the notion of neighboring databases to distinguish the impact of any particular individual's input on the output of a function. We likewise measure the maximum change in any function due to the replacement of a single tuple in the database, often calling it the sensitivity of the function.

DEFINITION 2.4 (SENSITIVITY). *Given a function f the sensitivity of f is $\max_{D' \sim D} |f(D) - f(D')|$ and is denoted by Δ_f .*

Differential privacy (DP) [23] protects the private information of individuals in the data by ensuring similar results for similar databases with high probability.

DEFINITION 2.5 (DIFFERENTIAL PRIVACY [23]). *Given a privacy budget $\epsilon > 0$, an algorithm $\mathcal{M}$ is said to satisfy ϵ -DP if for all $S \subseteq \text{Range}(\mathcal{M})$ and for all $D \sim D'$,*

$$Pr(\mathcal{M}(D) \in S) \leq e^\epsilon Pr(\mathcal{M}(D') \in S)$$

The Laplace mechanism [25] allows us to enforce DP by adding calibrated noise to query results and is a common building block in DP mechanisms.

DEFINITION 2.6 (LAPLACE MECHANISM [25]). *Given a database D , a function $f : \mathcal{D} \rightarrow \mathbb{R}$, and a privacy budget ϵ , the Laplace mechanism $\mathcal{M}_L$ returns $f(D) + v_q$, where $v_q \sim \text{Lap}(\Delta_f/\epsilon)$.*

THEOREM 2.7 (PRIVACY OF THE LAPLACE MECHANISM [26]). *Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be a query with the sensitivity Δ_f , and fix $\epsilon > 0$. The Laplace mechanism $\mathcal{M}_L(D) = f(D) + v$, with $v \sim \text{Lap}(\Delta_f/\epsilon)$, satisfies ϵ -differential privacy.*

3 PRIVATELY EVALUATING UNFAIRNESS

We formulate the problem of privately evaluating the unfairness of a database, starting with a discussion about measure requirements.

Measure desiderata. We define several requirements for private unfairness measures. The first two are inspired by the properties outlined for inconsistency measures in the context of logical integrity constraints [61], and the third one is specific to the DP setting. We aim to design an unfairness measure that takes a private database D , a set of fairness criteria $\mathcal{F}$, and a privacy budget ϵ , and returns a non-negative number, i.e., $\mathcal{U}(\mathcal{F}, D) \in [0, \infty)$. We want $\mathcal{U}$ to have the following desiderata:

Table 1: Properties of the discussed unfairness measures along with their computation and utility costs. Green rows denote the chosen measures and red rows denote measures that are unsuitable either due to relatively high sensitivity ($\mathcal{U}_{\text{MI}}$), unsatisfactory faithfulness to the original measure ($\mathcal{U}_{\text{MI}}^{\text{Bayes}}$), or high computation costs ($\mathcal{U}_{\text{R}}$). All measures satisfy the positivity and monotonicity properties outlined in Section 3, except $\mathcal{U}_{\text{MI}}^{\text{Bayes}}$ that violates positivity.

Motivation	Measure	Range for F	$\Delta_{\mathcal{U}}$ for F	$\Delta_{\mathcal{U}}$ for $\mathcal{F}$	Complexity	Error Bound
Attribute correlation	$\mathcal{U}_{\text{MI}}$	$[0, \infty)$	$O\left(\frac{\log n}{n}\right)$	$O\left(\frac{ \mathcal{F} \log n}{n}\right)$	-	-
	$\mathcal{U}_{\text{MI}}^{\text{Bayes}}$	$[-1, 0]$	$O\left(\frac{1}{n}\right)$	$O\left(\frac{ \mathcal{F} }{n}\right)$	-	-
	$\mathcal{U}_{\text{MI}}^{\text{TV}D}$	$[0, 2]$	$O\left(\frac{1}{n}\right)$	$O\left(\frac{ \mathcal{F} }{n}\right)$	$O(\mathcal{F} n)$	$\frac{8 \mathcal{F} }{n\epsilon}$
Distance to a fair database	$\mathcal{U}_{\text{R}}$	$[0, n]$	1	$ \mathcal{F} $	Exp. in n	-
	$\mathcal{U}_{\text{R}}^{\text{SAT}}$	$[0, n]$	2	$2 \mathcal{F} $	$O(\mathcal{F} (n^4 + \text{SAT}))$	$\frac{2 \mathcal{F} }{\epsilon}$
Contribution of the top- k tuples to unfairness	$\mathcal{U}_{\text{TC}}$	$[0, k]$	$O\left(\frac{4k}{n} + 2\right)$	$O\left(\mathcal{F} \left(\frac{4k}{n} + 2\right)\right)$	$O(\mathcal{F} n^3)$	$\frac{O(\mathcal{F} (\frac{4k}{n} + 2))}{\epsilon}$

- (1) **Positivity:** $\mathcal{U}(\mathcal{F}, D) \geq 0$ and $\mathcal{U}(\mathcal{F}, D) = 0$ iff D satisfies F for every $F \in \mathcal{F}$.
- (2) **Monotonicity:** $\mathcal{U}(\mathcal{F}, D) \leq \mathcal{U}(\mathcal{F}', D)$ if $\mathcal{F} \subseteq \mathcal{F}'$.
- (3) **Computability under DP:** We can efficiently compute $\mathcal{U}$ with an ϵ -DP algorithm with relatively small error.

The first two properties have already been suggested for inconsistency measures. Positivity was proposed as a fundamental property in [37, 70] and as an axiom in [62]. Monotonicity was proposed in different variations by previous work. Parisi and Grant [70] proposed measure monotonicity for database containment, i.e., $D \subseteq D'$ implies $\mathcal{U}(\mathcal{F}, D) \leq \mathcal{U}(\mathcal{F}, D')$, yet, in our case, database containment does not guarantee an increase of fairness, and in fact, may decrease it. Livshits et al. [61] suggested monotonicity for logical implication of integrity constraints, i.e., if $\Sigma \models \Sigma'$ for two sets of integrity constraints Σ, Σ' , then $\mathcal{U}(\Sigma', D) \leq \mathcal{U}(\Sigma, D)$. Since comparing fairness criteria (see Definition 2.2) through logical implication is not possible, we define a new notion of monotonicity for fairness criteria *set containment*. Intuitively, when more fairness criteria are used, the requirements from the database are stricter and, therefore, the value of the unfairness measure should only increase. The third property will be shown in two parts. First, showing that our measures have low sensitivity (Section 3) and then providing a DP algorithm for computing them (Section 4).

In the sequel, we will first define the measures for a single fairness criterion, $\mathcal{U}(F, D)$, and then extend them to a set of criteria as a sum $\mathcal{U}(\mathcal{F}, D) = \sum_{F \in \mathcal{F}} \mathcal{U}(F, D)$. We will then prove the properties for this extension. As we will see, measures and techniques from previous work in the non-private setting can be unsuitable for computation under DP, motivating our third desideratum and leading us to search for proxies or replacements.

Livshits et al. [61] lists two more properties that are specific for data repair settings with respect to integrity constraints, rather than our setting of measuring fairness. Specifically, the *continuity* property limits the effect of an ‘operation’ on the change in value of the measure, and the *progression* property states that there is always an ‘operation’ that can reduce the value of the measure. Herein, we consider a ‘static’ database, so there is no process for which continuity and progression are required.

The remainder of Section 3 presents our three measures, outlining their motivation and theoretical properties. We begin with $\mathcal{U}_{\text{MI}}^{\text{TV}D}$, which approximates mutual information between the attributes appearing in the fairness criteria (Section 3.1). We then introduce $\mathcal{U}_{\text{R}}^{\text{SAT}}$, a proxy for the minimal number of tuple insertions or deletions required to transform the dataset into one that satisfies the criteria (Section 3.2). Finally, we describe $\mathcal{U}_{\text{TC}}$, which reports the cumulative contribution of the top- k most influential tuples to the fairness criteria violation (Section 3.3). Table 1 provides a consolidated overview of the properties of all three measures.

3.1 Unfairness as Mutual Information

A standard way to measure dependence between variables is computing the mutual information (MI) between them [52, 64, 84]. Thus, for a fairness criterion $P \perp\!\!\!\perp Y \mid A$, we can measure the MI between P and Y given A to obtain our first measure notion. We abuse the notation from Definition 3.1 and denote the sum of (conditional) mutual information between the attributes in the fairness criteria contained in $\mathcal{F}$ by $MI_D(\mathcal{F})$, i.e., $MI_D(\mathcal{F}) = \sum_{F \in \mathcal{F}} MI_D(F)$.

DEFINITION 3.1 (MUTUAL INFORMATION UNFAIRNESS). *Given a database D and a fairness criterion of the form $F = P \perp\!\!\!\perp Y \mid A$ where A can be an empty set, the mutual information unfairness measure is defined as $\mathcal{U}_{\text{MI}}(F, D) = MI_D(P, Y \mid A)$.*

Previous work [84] has already shown that $\mathcal{U}_{\text{MI}}$ is not suitable as an accurate measure in the DP setting due to its relatively high sensitivity of $O(\log n/n)$. Such sensitivity means that Laplace noise of scale proportionate to $\log n/(\epsilon n)$ should be added to achieve ϵ -differential privacy. However, the range of $\mathcal{U}_{\text{MI}}$ is $[0, \infty)$, and the closer we get to F being satisfied (that is, the more independent the attributes in the F are), the smaller $\mathcal{U}_{\text{MI}}$ becomes. Thus, when F nearly holds and $\mathcal{U}_{\text{MI}}$ is close to 0, the added Laplace noise can severely distort the original $\mathcal{U}_{\text{MI}}$ value, making it unusable. The range of $\mathcal{U}_{\text{MI}}$ also poses a problem. Since it is unbounded, it is difficult to interpret its values and distinguish which values indicate low dependence score between the sensitive and outcome attributes, and which values indicate high dependence score. Therefore, we need to find an alternative with lower sensitivity.

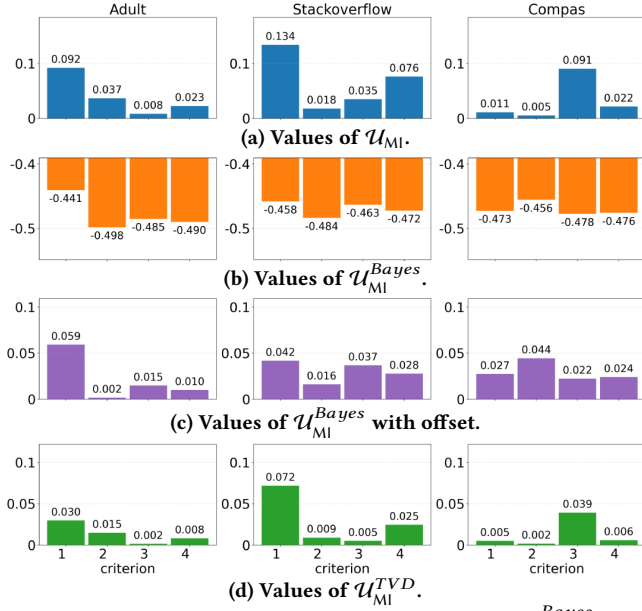


Figure 3: Demonstrating the faithfulness of $\mathcal{U}_{MI}^{Bayes}$ and $\mathcal{U}_{MI}^{TVD}$ to $\mathcal{U}_{MI}$ without privacy considerations.

To mitigate these issues, one can use a proxy function for $\mathcal{U}_{MI}$ that has low sensitivity and a bounded range. A possible proxy function, presented in [84] for a single fairness criterion, is $\mathcal{U}_{MI}^{Bayes}(F, D) = -\frac{1}{2} \min_{P \in \mathcal{P}} \|Pr^\diamond[P, Y | A] - Pr[P, Y | A]\|$, where $Pr^\diamond[P, Y | A]$ is the maximum joint distribution defined as the one that maximizes the mutual information between P and Y , given A . Zhang et al. [84] also showed that the sensitivity of $\mathcal{U}_{MI}^{Bayes}$ is $\frac{1}{n}$ for an unconditional fairness criterion, and is therefore suitable for DP.

However, $\mathcal{U}_{MI}^{Bayes}$ lacks the desired positivity property. Furthermore, the relation between $\mathcal{U}_{MI}$ and $\mathcal{U}_{MI}^{Bayes}$ can be difficult to visually comprehend. Figure 3 demonstrates the faithfulness between $\mathcal{U}_{MI}$ (Figure 3a) and $\mathcal{U}_{MI}^{Bayes}$ (Figure 3b) for the Adult, StackOverflow survey, and Compas datasets (see Footnote 1) with fairness criteria from Table 2, numbered for each dataset. We also plotted $\mathcal{U}_{MI}^{Bayes}$ with an offset of $\frac{1}{2}$ (Figure 3c) to make its values positive so the reader could easily see the variation between $\mathcal{U}_{MI}$ and $\mathcal{U}_{MI}^{Bayes}$ across all datasets and criteria.

This faithfulness gap motivates us to define a novel proxy function for $\mathcal{U}_{MI}$ that has low sensitivity and high correlation with $\mathcal{U}_{MI}$ both theoretically and in practice. This proxy measure is based on the concepts of Total Variation Distance (TVD), that is defined as $TVD(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|$ for two distributions P and Q .

DEFINITION 3.2 ($\mathcal{U}_{MI}^{TVD}$ AS A PROXY FOR $\mathcal{U}_{MI}$). Given a database D and a fairness criterion of the form $F = P \perp\!\!\!\perp Y \mid A$ where A can be an empty set, the $\mathcal{U}_{MI}^{TVD}$ proxy for $\mathcal{U}_{MI}$ is defined as:

$$\mathcal{U}_{MI}^{TVD}(P \perp\!\!\!\perp Y \mid A, D) = 2 \cdot (TVD(Pr_D(P, Y \mid A), Pr_D(P \mid A)Pr_D(Y \mid A)))^2$$

We first show that $\mathcal{U}_{MI}^{TVD}$ is a bounded approximation of $\mathcal{U}_{MI}$ and further show that it leads to a better approximation of $\mathcal{U}_{MI}$ in practical scenarios.

PROPOSITION 3.3 ($\mathcal{U}_{MI}^{TVD}$ IS BOUNDED BY $\mathcal{U}_{MI}$). Let D be a database such that the schema of D is a superset of $P \cup Y$. Given a fairness constraint $P \perp\!\!\!\perp Y$, the following holds: $\alpha \cdot \mathcal{U}_{MI}(P, Y) \leq \mathcal{U}_{MI}^{TVD}(P \perp\!\!\!\perp Y, D) \leq \mathcal{U}_{MI}(P, Y)$ where $X = Pr_D(P, Y)$, $Y = Pr_D(P)Pr_D(Y)$, and $\alpha = \min_{z \in Z, Y(z) > 0} Y(z)$.

The proofs for all propositions and lemmas can be found in Appendix A.

Though the lower bound could be loose when α is small given a skewed dataset, in practice $\mathcal{U}_{MI}^{TVD}$ approximates $\mathcal{U}_{MI}$ tightly. Figure 3d shows the values of $\mathcal{U}_{MI}^{TVD}$ as a comparison to $\mathcal{U}_{MI}$, $\mathcal{U}_{MI}^{Bayes}$, and its offset version. The trend of $\mathcal{U}_{MI}^{TVD}$ values emulates the trend of $\mathcal{U}_{MI}$ values better than the other measures.

We now show that $\mathcal{U}_{MI}^{TVD}$ satisfies the desired properties. The first two properties can be shown directly, while computability under DP will be proven in a two part fashion: (1) bounding the sensitivity and range of the measure in the following proposition and (2) providing an ϵ -DP algorithm for computing it in Section 4.

PROPOSITION 3.4 ($\mathcal{U}_{MI}^{TVD}$ SATISFIES THE DESIRED PROPERTIES).

- (1) $\mathcal{U}_{MI}^{TVD}$ satisfies the Positivity property.
- (2) $\mathcal{U}_{MI}^{TVD}$ satisfies the Monotonicity property.
- (3) The range of $\mathcal{U}_{MI}^{TVD}$ is $[0, 2|\mathcal{F}|]$ for a set of criteria $\mathcal{F}$.
- (4) The sensitivity of $\mathcal{U}_{MI}^{TVD}$ is $\frac{16|\mathcal{F}|}{n}$ for a set of criteria $\mathcal{F}$ and a database of size n .

3.2 Unfairness as Data Repair Cost

Inspired by the field of data repair [3, 9, 16, 33, 60, 74] and previous work on inconsistency measures for integrity constraints [61, 70], including a recent work that allows for their DP computation [66], we use a similar idea to define another unfairness measure.

DEFINITION 3.5 (DATA REPAIR UNFAIRNESS). Given a database D and a fairness criterion of the form $F = P \perp\!\!\!\perp Y \mid A$ where A can be an empty set, the repair unfairness measure is defined as $\mathcal{U}_R(F, D) = |D \dot{-} D_R|$, where D_R is the database with the smallest number of removed and added tuples with respect to D that satisfies F and $\dot{-}$ is the symmetric difference.

It follows that $\mathcal{U}_R$ satisfies the Positivity and Monotonicity properties, its range is $[0, n|\mathcal{F}|]$, and its sensitivity is $|\mathcal{F}|$. However, the problem of finding a minimum data repair is known to be a particularly challenging one, as we shall next elaborate.

Computational challenge. The problem of data repair has been thoroughly studied with different intervention models, such as tuples deletions [35, 59], value update [16, 33, 72], and combinations of tuple additions and deletions [74]. Its computational hardness is well established, even under tuple deletion alone [59]. Although our goal is to measure the size of the repair with deletions and additions and not to find the repair itself, the two problems are computationally equivalent. To address this challenge, in Section 4, we adapt an algorithm from previous work [74] that reduces the problem of computing $\mathcal{U}_R$ to the Max-3SAT problem, and then uses a SAT solver to obtain a repair. Therefore, for $\mathcal{U}_R$ we resort to a proxy measure called $\mathcal{U}_R^{SAT}$ that is based on prior work [74], not for privacy purposes, but for computational purposes.

Review of the approach from [74]. To overcome the tractability issue, Salimi et al. [74] have proposed a reduction from the problem of computing an optimal repair based on a fairness criterion to the problem of weighted MaxSAT, which can then be solved by a SAT solver. Assume that we wish to repair the data, so that the fairness criterion $P \perp\!\!\!\perp Y \mid A$ holds, where the set of attributes of D is $\mathcal{A} = \{P, Y, A\}$.²

Define the self-join database $D_{sj} = \Pi_{P,A}(D) \bowtie \Pi_{Y,A}(D)$. Then, any minimal repair D' of D satisfies $D' \subseteq D_{sj}$. The reduction converts the tuples in D_{sj} to a CNF formula by adding a clause x_t for every $t \in D$ and a clause $\neg x_t$ for every $t \in D_{sj} \setminus D$. These are the ‘soft clauses’ of the formula. It also adds the clauses $\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3}$ for tuples of the form $t_1[p_1, y_1, a]$, $t_2[p_2, y_2, a]$ and $t_3[p_1, y_2, a]$. Intuitively, t_1 and t_2 are the lineage of t_3 . Thus, these are the ‘hard clauses’ of the formula, i.e., they must be satisfied by the obtained assignment. Salimi et al. [74] uses a SAT solver to find an assignment that maximizes the additive weight of the satisfied soft clauses of the constructed CNF formula. The assignment directly corresponds to the tuples that have to be removed and added to satisfy the fairness criterion. Next, we show how we utilize and extend this approach to define $\mathcal{U}_R^{SAT}$, which is our proxy for $\mathcal{U}_R$.

3.2.1 Defining $\mathcal{U}_R^{SAT}$. Following previous work [74], we define $\mathcal{U}_R^{SAT}$ in terms of an assignment to the relevant CNF formula. We begin by defining this CNF formula that D_{sj} is mapped to.

DEFINITION 3.6 (CNF FORMULA FOR D_{sj} [74]). *Given a database D whose schema contains $P \cup Y \cup A$, and a fairness criterion $F = P \perp\!\!\!\perp Y \mid A$, the self-join database CNF formula is defined as*

$$\varphi(D, D_{sj}) = \mathcal{H}(D_{sj}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj} \setminus D} (\neg x_t),$$

where each x_t represents a soft clause for a tuple t and $\mathcal{H}(D_{sj})$ is the set of hard clauses constructed as follows. Let $C(P_1, Y_1, P_2, Y_2, A) := D_{sj}(P_1, Y_1, A) \wedge D_{sj}(P_2, Y_2, A)$. That is, C is the set of all tuples formed by joining D_{sj} with itself on the attribute A , pairing tuples with the same A -values and extracting the relevant attributes. For each tuple $t \in C$, a clause of the form $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$ is added to $\mathcal{H}(D_{sj})$, where $t_1 = (P_1, Y_1, A)$, $t_2 = (P_2, Y_2, A)$, $t_3 = (P_1, Y_2, A)$.

We treat the soft clauses $\bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj} \setminus D} (\neg x_t)$, as a bag, meaning that soft clauses for two identical tuples (except their IDs) will both appear in $\varphi(D, D_{sj})$.

A feasible assignment for a CNF formula $\varphi(D, D_{sj})$ is an assignment, α , that satisfies all the hard clauses $\mathcal{H}(D_{sj})$.

We now define the repair cost of a database based on an assignment from Definition 3.6. We will then link this notion with the notion of data repair by means of tuple deletions and additions.

DEFINITION 3.7 (REPAIR COST UNDER ASSIGNMENT). *Given a database D , a fairness criterion F , a CNF formula $\varphi(D, D_{sj})$ and α a feasible assignment for φ , let D_R denote the set of tuples $t \in D_{sj}$ such that x_t is assigned True under the assignment α . We define the cost of repairing D under the assignment α as $\text{dist}(\varphi(D, D_{sj}), \alpha) = |D \div D_R|$.*

²In practice, the database schema may contain additional attributes that do not appear in the fairness criterion. We therefore compute the repair on the projection to the criterion’s attributes and, when we lift this repaired projection back to a full repair of the original schema, we keep all other attributes as in the original database. Since these additional attributes are unchanged, all propositions that we prove in the sequel with the reduced schema assumption carry over to the full schema.

That is, the size of the set of tuples in D whose corresponding variables are assigned to False, or are not present in D and assigned to True.

The following lemma discusses the direct translation between the number of soft clauses satisfied by an assignment to the CNF formula and the number of changes in the database required for it to satisfy the fairness criterion. In particular, a larger number of satisfied clauses means a smaller required change in the original database.

LEMMA 3.8. *Given a database D and a fairness criterion, let D_{sj} be the self-join database. Let $\varphi(D, D_{sj})$ be a CNF formula defined according to Definition 3.6 and let α_1 and α_2 be two feasible assignments for φ . Finally, let $\text{dist}(\varphi(D, D_{sj}), \alpha_1)$ and $\text{dist}(\varphi(D, D_{sj}), \alpha_2)$ be defined according to Definition 3.7. If α_1 satisfies more soft clauses than α_2 , then: $\text{dist}(\varphi(D, D_{sj}), \alpha_1) < \text{dist}(\varphi(D, D_{sj}), \alpha_2)$.*

By applying Lemma 3.8, we can define a proxy for $\mathcal{U}_R$, $\mathcal{U}_R^{SAT}$, as the repair with the minimum cost based on Definition 3.7.

DEFINITION 3.9 ($\mathcal{U}_R^{SAT}$ AS A PROXY FOR $\mathcal{U}_R$). *Given a database D , a fairness criterion F , and the CNF formula $\varphi(D, D_{sj})$, the cost of an optimal repair of D through $\varphi(D, D_{sj})$ is*

$$\mathcal{U}_R^{SAT}(F, D) := \min_{\alpha \models \mathcal{H}(D_{sj})} \text{dist}(\varphi(D, D_{sj}), \alpha)$$

We detail the properties of $\mathcal{U}_R^{SAT}$, showing that it satisfies the required desiderata and its bounded sensitivity relative to its range.

PROPOSITION 3.10 ($\mathcal{U}_R^{SAT}$ SATISFIES THE DESIRED PROPERTIES).

- (1) $\mathcal{U}_R^{SAT}$ satisfies the Positivity property.
- (2) $\mathcal{U}_R^{SAT}$ satisfies the Monotonicity property.
- (3) The range of $\mathcal{U}_R^{SAT}$ is $[0, |\mathcal{F}|n]$ for a set of criteria $\mathcal{F}$.
- (4) The sensitivity of $\mathcal{U}_R^{SAT}$ is $2|\mathcal{F}|$ for a set of criteria $\mathcal{F}$.

3.3 Unfairness as Top Contributions

Previous work has measured tuple contributions as both measuring inconsistency [61] and as means of explaining results [20, 57, 58, 65]. We adapt this notion to define an unfairness measure that distinguishes between databases that have a relatively small number of outliers that violate fairness criteria, and those that have a more ‘uniform distribution’ of fairness violating tuples. First, we define the notion of marginal difference for a fairness criterion $F = P \perp\!\!\!\perp Y \mid A$ where A can be absent, as follows:

$$\begin{aligned} \mathcal{MD}(F, D, t) = & \Pr_D(A = t[A]) \left| \Pr_D(P = t[P], Y = t[Y] \mid A = t[A]) \right. \\ & \left. - \Pr_D(P = t[P] \mid A = t[A]) \Pr_D(Y = t[Y] \mid A = t[A]) \right|, \end{aligned}$$

where $\Pr_D(A = t[A]) = 1$ when A is absent.

We now define the $\mathcal{U}_{TC}$ measure using the $\mathcal{MD}$ notion.

DEFINITION 3.11 (TOP- k TUPLE CONTRIBUTION UNFAIRNESS). *Given a database D , a natural number k , and a fairness criterion of the form $F = P \perp\!\!\!\perp Y \mid A$, define top- k as the set of top- k tuples based on their $\mathcal{MD}$ values. We assume that each tuple in the database has a unique ID, so that tuples with the same values but different IDs can both appear in the top- k . Then, the tuple contribution unfairness measure is defined as $\mathcal{U}_{TC}(F, D) = \sum_{t \in \text{top-}k} \mathcal{MD}(F, D, t)$.*

Definition 3.11 resembles Definition 3.2 in terms of being a cumulative sum of the residual difference of tuples. However, it differs in cases where we have two databases with similar sum of individual contributions and different distribution of these contributions, e.g., when a few outliers in one of the databases have significantly larger contributions than the rest of the tuples (a ‘long tail’). We provide a sample experiment in Appendix A to demonstrate the practical distinction between $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ and $\mathcal{U}_{\text{TC}}$.

PROPOSITION 3.12 ($\mathcal{U}_{\text{TC}}$ SATISFIES THE DESIRED PROPERTIES). *Given a natural number k , the following holds for $\mathcal{U}_{\text{TC}}$:*

- (1) $\mathcal{U}_{\text{TC}}$ satisfies the Positivity property.
- (2) $\mathcal{U}_{\text{TC}}$ satisfies the Monotonicity property.
- (3) The range of $\mathcal{U}_{\text{TC}}$ is $[0, k|\mathcal{F}|]$ for a set of criteria $\mathcal{F}$.
- (4) The sensitivity of $\mathcal{U}_{\text{TC}}$ is $|\mathcal{F}| \left(\frac{3k}{n} + 2 \right)$ for a set of unconditional $\mathcal{F}$, and $|\mathcal{F}| \left(\max_{a \in A | A \in \mathcal{F}} \left| \left\{ t \in D \mid t[A] = a \right\} \right| \right) \frac{4k}{n} + 2$ for a set of conditional $\mathcal{F}$, where $A \in \mathcal{F}$ is the set of admissible attributes participating in a fairness criterion in $\mathcal{F}$.

4 PRIVATE MEASURE COMPUTATION

We now describe the algorithms that compute the three unfairness measures while complying with DP, showing the third desideratum from Section 3.

4.1 Computing $\mathcal{U}_{\text{MI}}^{\text{TVD}}$

The pseudocode for computing $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ is summarized in Algorithm 1. In line 1, the algorithm initializes to zero the variable that will accumulate $\mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D)$ for all F in the set of criteria $\mathcal{F}$. For each fairness criterion of the form $F = P \perp\!\!\!\perp Y \mid A$ in the set of criteria $\mathcal{F}$, in lines 3-5 the algorithm computes empirical probabilities derived from the dataset D : the conditional joint probability $Pr_D(P, Y \mid A)$ and the conditional marginal probabilities $Pr_D(P \mid A)$ and $Pr_D(Y \mid A)$. In case A is not given, we assume that $Pr_D(P, Y \mid A)$ is equal to $Pr_D(P, Y)$, $Pr_D(P \mid A)$ is equal to $Pr_D(P)$, and $Pr_D(Y \mid A)$ is equal to $Pr_D(Y)$.

Then, the algorithm splits into two cases: in the unconditional case (when A is not given), it computes the TVD as per the usual formula in line 8, and in the conditional case for every value a of the admissible attributes, the algorithm first computes $Pr_D(A = a)$ for every a in A in line 10, and then computes the sum of the conditional TVDs weighted by these probabilities in line 12.

Finally, the algorithm computes $\mathcal{U}_{\text{MI}}^{\text{TVD}}(F, D)$ as $2 \cdot \text{TVD}^2$ (Definition 3.2) and updates the cumulative sum with this value in line 13. After computing $\mathcal{U}_{\text{MI}}^{\text{TVD}}$, the algorithm applies the Laplace mechanism in line 14 by adding noise according to the sensitivity $\frac{16|\mathcal{F}|}{n}$ (Item 4 in Proposition 3.4), and the privacy budget ϵ . The resulting value $\widehat{\mathcal{U}}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D)$ is then returned.

We next show that Algorithm 1 satisfies DP with bounded error, and thus $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ satisfies the third property in Section 3.

PROPOSITION 4.1 (DP AND ERROR BOUND OF ALGORITHM 1).

- (1) Algorithm 1 is ϵ -DP.
- (2) For a database D and a set of fairness criteria $\mathcal{F}$, Algorithm 1 returns $\widehat{\mathcal{U}}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D)$ such that for any $\epsilon > 0$ and $0 < \delta < 1$,

Algorithm 1: Compute $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ under DP

Input: Database D ; set of fairness criteria $\mathcal{F}$; privacy budget ϵ

Output: $\widehat{\mathcal{U}}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D)$

```

1  $\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) \leftarrow 0$ ;
2 foreach  $F = (P \perp\!\!\!\perp Y \mid A) \in \mathcal{F}$  do
3   Compute  $Pr_D(P = p, Y = y \mid A = a)$ ;
4   Compute  $Pr_D(P = p \mid A = a)$ ;
5   Compute  $Pr_D(Y = y \mid A = a)$ ;
6   /* Unconditional criterion */
7   if  $A$  is  $\emptyset$  then
8      $\text{TVD} \leftarrow \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_D(P = p, Y = y) - Pr_D(P = p) Pr_D(Y = y)|$ ;
9   /* Conditional criterion */
10  else
11    Compute  $Pr_D(A = a)$  for every  $a \in A$ ;
12     $\text{TVD} \leftarrow \sum_{a \in A} Pr_D(A = a) \cdot \left( \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_D(P = p, Y = y \mid A = a) - Pr_D(P = p \mid A = a) Pr_D(Y = y \mid A = a)| \right)$ ;
13   $\mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) \leftarrow \mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) + 2 \cdot \text{TVD}^2$ ;
14 Draw  $\eta \sim \text{Lap}\left(0, \frac{16|\mathcal{F}|}{n\epsilon}\right)$ ;
15  $\widehat{\mathcal{U}}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) \leftarrow \mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) + \eta$ ;
16 return  $\widehat{\mathcal{U}}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D)$ ;

```

with probability at least $1 - \delta$ it holds that

$$\mathbb{E} \left[\left| \widehat{\mathcal{U}}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) - \mathcal{U}_{\text{MI}}^{\text{TVD}}(\mathcal{F}, D) \right| \right] = \frac{16|\mathcal{F}|}{n\epsilon}$$

Complexity of Algorithm 1. The complexity of Algorithm 1 is $O(|\mathcal{F}|n)$. For every F , the algorithm first computes the joint and the marginal empirical probabilities. Assuming the counts are stored in hash tables, so every lookup and update takes $O(1)$ time, this whole part takes $O(n)$ time. Then, the algorithm computes the TVD and divides into two cases. In the unconditional case it goes over all the tuples in D once when calculating the sum, so the whole computation takes $O(n)$ time. And in the conditional case it goes over all the values a in A and calculates the sum from the unconditional case for it. This computation takes $O(n + n) = O(n)$ time. Updating the cumulating sum takes $O(1)$ time. Finally, after the algorithm finishes going over all F in $\mathcal{F}$, it adds Laplace noise to the resulting cumulative sum and returns it, which takes $O(1)$.

4.2 Computing $\mathcal{U}_{\text{R}}^{\text{SAT}}$

Since the scale of sufficient noise has been determined in Section 3.2.1, we can now describe Algorithm 2 that combines the conversion to a weighted CNF formula as in [74], uses a SAT solver to obtain an assignment to the weighted MaxSAT problem, and returns the result while satisfying DP.

In line 1, the algorithm initializes the variable that will accumulate $\mathcal{U}_{\text{R}}^{\text{SAT}}(F, D)$ for all F in the set $\mathcal{F}$. For each fairness criterion $F = P \perp\!\!\!\perp Y \mid A$ in the set $\mathcal{F}$, the code block 3 runs the algorithm from [74]. This block first initializes φ to an empty set, and then

adds soft and hard clauses as in definition 3.6. Then, the algorithm calls a SAT solver in line 4 to obtain an assignment α that satisfies all the hard clauses and maximizes the number of satisfied soft clauses in φ . From this assignment, the algorithm constructs the repaired database in line 5, which has all the tuples that α assigns True. In line 5 it computes $\mathcal{U}_R^{SAT}(F, D)$ according to Definition 3.9, and updates the cumulative sum with this value.

Finally, the algorithm applies the Laplace mechanism in line 7 and adds noise according to the sensitivity $2|\mathcal{F}|$ (see Item 4 in Proposition 3.10). The noisy version of $\mathcal{U}_R^{SAT}$ is then returned.

Algorithm 2: Compute $\mathcal{U}_R^{SAT}$ under DP

Input: Database D ; set of fairness criteria $\mathcal{F}$; privacy budget ϵ

Output: $\widehat{\mathcal{U}}_R^{SAT}(\mathcal{F}, D)$

```

1  $\mathcal{U}_R^{SAT}(\mathcal{F}, D) \leftarrow 0$ ;
2 foreach  $F = (P \perp Y \mid A) \in \mathcal{F}$  do
    /* Algorithm from [74] */
     $D_{sj}(P_1, Y_2, A) \leftarrow D(P_1, Y_1, A) \bowtie D(P_2, Y_2, A)$ ;
     $\varphi \leftarrow \emptyset$ ;
    foreach  $t \in D_{sj}$  do
        if  $t \in D$  then
            | Add the soft clause  $x_t$  to  $\varphi$ ;
        if  $t \notin D$  then
            | Add the soft clause  $\neg x_t$  to  $\varphi$ ;
     $C(P_1, Y_1, P_2, Y_2, A) \leftarrow D_{sj}(P_1, Y_1, A) \wedge D_{sj}(P_2, Y_2, A)$ ;
    foreach  $t \in C$  do
         $t_1 \leftarrow t(P_1, Y_1, A)$ ,
         $t_2 \leftarrow t(P_2, Y_2, A)$ ,
         $t_3 \leftarrow t(P_1, Y_2, A)$ ;
        | Add the hard clause  $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$  to  $\varphi$ ;
3  $\alpha \leftarrow \text{Solver}(\varphi)$ ;
4  $D_R \leftarrow \{t \mid \alpha(x_t) = \text{True}\}$ 
5  $\mathcal{U}_R^{SAT}(\mathcal{F}, D) \leftarrow \mathcal{U}_R^{SAT}(\mathcal{F}, D) + |D \div D_R|$ ;
6 Draw  $\eta \sim \text{Lap}\left(0, \frac{2|\mathcal{F}|}{\epsilon}\right)$ ;
7  $\widehat{\mathcal{U}}_R^{SAT}(\mathcal{F}, D) \leftarrow \mathcal{U}_R^{SAT}(\mathcal{F}, D) + \eta$ ;
8 return  $\widehat{\mathcal{U}}_R^{SAT}(\mathcal{F}, D)$ ;
    
```

PROPOSITION 4.2 (DP AND ERROR BOUND ON ALGORITHM 2).

- (1) Algorithm 2 is ϵ -DP.
- (2) For a database D and a set of fairness criteria $\mathcal{F}$, Algorithm 2 returns $\widehat{\mathcal{U}}_R^{SAT}(\mathcal{F}, D)$ such that for any $\epsilon > 0$ and $0 < \delta < 1$, with probability at least $1 - \delta$ it holds that $\mathbb{E} \left[\left| \widehat{\mathcal{U}}_R^{SAT}(\mathcal{F}, D) - \mathcal{U}_R^{SAT}(\mathcal{F}, D) \right| \right] = \frac{2|\mathcal{F}|}{\epsilon}$.

Complexity of Algorithm 2. The complexity of Algorithm 2 is $O(|\mathcal{F}|(n^4 + \text{SAT}))$, where $O(\text{SAT})$ is the complexity of the SAT solver, due to the following analysis. For every F , the algorithm first computes the self-join D_{sj} of D on A and goes over it to add the soft clauses to φ . It takes $O(n)$ time to calculate the per- a projection sets,

so it takes $O(n^2)$ overall to perform the join and iterate over it to compute the soft clauses. Then the algorithm computes the self-join C of D_{sj} on A . Since each copy of D_{sj} contains $O(|P_a| \times |Y_a|)$ tuples for any a , this takes $O\left(\sum_{a \in A} |P_a|^2 |Y_a|^2\right) = O(n^4)$ time. Finally, the algorithm runs a solver to get an assignment for the constructed φ , computes the symmetric difference and updates the cumulative sum. This whole part takes $O(\text{SAT} + n)$. After the algorithm finishes iterating over all $F \in \mathcal{F}$, it adds Laplace noise to the resulting sum and returns it, which takes $O(1)$ time.

4.3 Computing $\mathcal{U}_{TC}$

We describe the pseudocode in Algorithm 3 for computing $\mathcal{U}_{TC}$. In line 1, the algorithm initializes to zero the variable that will accumulate $\mathcal{U}_{TC}(F, D)$ for all F in the set of criteria $\mathcal{F}$. For each fairness criterion of the form $F = P \perp Y \mid A$ in the set of criteria $\mathcal{F}$, in lines 3-5 the algorithm computes empirical probabilities derived from the dataset D : the conditional joint probability $Pr_D(P, Y \mid A)$ and the conditional marginal probabilities $Pr_D(P \mid A)$ and $Pr_D(Y \mid A)$. In case A is not given, we assume that $Pr_D(P, Y \mid A)$ is equal to $Pr_D(P, Y)$, $Pr_D(P \mid A)$ is equal to $Pr_D(P)$, and $Pr_D(Y \mid A)$ is equal to $Pr_D(Y)$. Then, the algorithm iterates over every tuple t in D and in line 7 it computes the marginal difference $\mathcal{MD}(F, D, t)$ (defined in Section 3.3). The resulting value quantifies the contribution of each tuple t to the deviation of the observed joint probability from the joint probability in the case when the independence defined by F would hold. After computing marginal differences for all tuples in the dataset, the algorithm sorts them in descending order in line 8. Then in line 9 it adds the sum of the k largest marginal differences to the global value $\mathcal{U}_{TC}(\mathcal{F}, D)$. In other words, it adds to the total score, across all criteria in $\mathcal{F}$, how much the independence defined by a single criterion is violated in the dataset by the k tuples that violate it the most. The algorithm performs all the previous stages for every F in $\mathcal{F}$. Finally, after going over all of them, the algorithm applies the Laplace mechanism in line 14 and adds noise according to the sensitivity $|\mathcal{F}| \left(\max_{a \in A} |A \in \mathcal{F} \mid \frac{4k}{| \{t \in D \mid t[A]=a\} |} | + 2 \right)$ if for one of $F \in \mathcal{F}$, A is given (conditional case), and $|\mathcal{F}| \left(\frac{3k}{n} + 2 \right)$ otherwise (unconditional case). (Item 4 in Proposition 3.12). The resulting value is then returned as the DP version of the contribution score.

PROPOSITION 4.3 (DP AND ERROR BOUND OF ALGORITHM 3).

- (1) Algorithm 3 is ϵ -DP.
- (2) For a database D , Algorithm 3 returns $\widehat{\mathcal{U}}_{TC}(\mathcal{F}, D)$ such that for any $\epsilon > 0$ and $0 < \delta < 1$, with probability at least $1 - \delta$ it holds that

$$\mathbb{E} \left[\left| \widehat{\mathcal{U}}_{TC}(\mathcal{F}, D) - \mathcal{U}_{TC}(\mathcal{F}, D) \right| \right] = \frac{|\mathcal{F}| \left(\frac{3k}{n} + 2 \right)}{\epsilon}$$

for a set of unconditional fairness criteria $\mathcal{F}$, and

$$\mathbb{E} \left[\left| \widehat{\mathcal{U}}_{TC}(\mathcal{F}, D) - \mathcal{U}_{TC}(\mathcal{F}, D) \right| \right] = \frac{|\mathcal{F}| \left(\max_{a \in A} |A \in \mathcal{F} \mid \frac{4k}{| \{t \in D \mid t[A]=a\} |} | + 2 \right)}{\epsilon}$$

for a set of conditional fairness criteria.

Complexity of Algorithm 3. The complexity of Algorithm 3 is $O(|\mathcal{F}|n^3)$. For every F , the algorithm first computes the joint and the marginal empirical probabilities, which takes $O(|P||Y||A| + n) = O(n^3)$ time. Then, the algorithm computes the marginal differences, taking $O(n)$ time. Sorting of the marginal differences for all the

Algorithm 3: Compute the ϵ -DP version of $\mathcal{U}_{TC}$

Input: Database D ; set of fairness criteria $\mathcal{F}$; accuracy parameter $k \in \mathbb{N}$; privacy budget ϵ

Output: $\widetilde{\mathcal{U}}_{TC}(\mathcal{F}, D)$

```

1  $\mathcal{U}_{TC}(\mathcal{F}, D) \leftarrow 0$ ;
2 foreach  $F = (P \perp\!\!\!\perp Y \mid A) \in \mathcal{F}$  do
3   Compute  $Pr_D[P = p, Y = y \mid A = a]$ ;
4   Compute  $Pr_D[P = p \mid A = a]$ ;
5   Compute  $Pr_D[Y = y \mid A = a]$ ;
6   foreach  $t \in D$  do
7      $\mathcal{MD}_D(F, t) \leftarrow Pr_D[A = a] \left| Pr_D[P = p, Y = y \mid A = a] - Pr_D[P = p \mid A = a] Pr_D[Y = y \mid A = a] \right|$ ;
8   Sort  $\mathcal{MD}_D(F, t)$  for all  $t \in D$  in descending order;
9    $\mathcal{U}_{TC}(\mathcal{F}, D) \leftarrow \mathcal{U}_{TC}(\mathcal{F}, D) + \sum_{t \in \text{top-}k} \mathcal{MD}_D(F, t)$ ;
10 if  $\exists F = (P, Y \mid A) \in \mathcal{F}$  such that  $A$  is given then
11    $\Delta \leftarrow \max_{a \in A} \max_{A \in \mathcal{F}} \frac{4k}{|\{t \in D \mid t[A] = a\}|} + 2$ ;
12 else
13    $\Delta \leftarrow \frac{3k}{n} + 2$ ;
14 Draw  $\eta \sim \text{Lap}\left(0, \frac{|\mathcal{F}| \cdot \Delta}{\epsilon}\right)$ ;
15  $\widetilde{\mathcal{U}}_{TC}(\mathcal{F}, D) \leftarrow \mathcal{U}_{TC}(\mathcal{F}, D) + \eta$ ;
16 return  $\widetilde{\mathcal{U}}_{TC}(\mathcal{F}, D)$ ;

```

tuples and finding the k largest ones takes $O(n \log k)$ time if we are using a size- k heap. Summing the k largest marginal differences takes $O(k)$ time and updating the cumulative sum takes $O(1)$ time. Therefore, this whole part takes $O(n \log k + k) = O(n \log n)$. After going over all F in $\mathcal{F}$, it adds Laplace noise to the resulting cumulative sum and returns it, which takes $O(1)$ time.

5 EXPERIMENTS

We have experimentally studied our measures and the algorithms computing them by answering the following questions: (1) What is the ability of the measures to detect different levels of unfairness? (2) How do our algorithms scale? (3) What is the effect of the privacy budget on measure accuracy? (4) How do the measures perform in terms of faithfulness and under parameter variation?

5.1 Experimental Setup

All algorithms and experiments were implemented in Python using standard packages such as `networkx`, `numpy`, `pandas`, and `z3-solver`. The experiments ran on a machine with Intel Core i5-12400F. For each data point in every experiment, we performed 10 independent repetitions, unless stated otherwise, and averaged the result. If one of these repetitions took longer than two hours to complete, we did not include this data point in the graph. For experiments with fixed data size, we randomly sampled up to 100K tuples from each dataset, or used the entire dataset if it is smaller.

Datasets. We used the following five datasets in our experiments.

- **Adult:** A dataset downloaded from ³, containing 15 attributes and 48,843 tuples. It contains personal data about individuals, including

³<https://archive.ics.uci.edu/dataset/2/adult>

Table 2: Fairness criteria per dataset.

Dataset	Fairness criteria
Adult	(1) education-num $\perp\!\!\!\perp$ income>50K (2) sex $\perp\!\!\!\perp$ income>50K (3) race $\perp\!\!\!\perp$ income>50K (4) sex $\perp\!\!\!\perp$ income>50K hours-per-week
IPUMS-CPS	(1) HEALTH $\perp\!\!\!\perp$ INCTOT EDUC (2) HEALTH $\perp\!\!\!\perp$ OCC EDUC (3) HEALTH $\perp\!\!\!\perp$ MARST AGE (4) HEALTH $\perp\!\!\!\perp$ INCTOT AGE
Stackoverflow	(1) Country $\perp\!\!\!\perp$ RemoteWork Employment (2) Age $\perp\!\!\!\perp$ PurchaseInfluence OrgSize (3) Country $\perp\!\!\!\perp$ MainBranch YearsCodePro (4) Age $\perp\!\!\!\perp$ MainBranch EdLevel
Compas	(1) race $\perp\!\!\!\perp$ is_recid age_cat (2) sex $\perp\!\!\!\perp$ is_recid priors_count (3) race $\perp\!\!\!\perp$ decile_score c_charge_degree (4) sex $\perp\!\!\!\perp$ v_decile_score age_cat
Healthcare	(1) race $\perp\!\!\!\perp$ complications age_group (2) smoker $\perp\!\!\!\perp$ complications age_group (3) race $\perp\!\!\!\perp$ income county (4) smoker $\perp\!\!\!\perp$ income num_children

their sex, race, marital status, whether their income is over 50K, etc. Some attributes were renamed for clarity or to match similar ones in other datasets (e.g., income was renamed to income>50K).

- **IPUMS-CPS:** A population survey dataset published by the U.S. Census Bureau [30] and downloaded from ⁴, containing 8 attributes and 1,048,576 tuples, with data from 2011 till 2019. The dataset, like the Adult dataset, contains personal data about individuals.

- **Stackoverflow** ⁵: A dataset containing 114 attributes and 65,438 tuples. The dataset contains the results of the Stackoverflow developers survey for 2024, and includes languages they use, where they learned to code, information on the companies they work at, etc.

- **Compas** ⁶: A dataset containing 47 attributes and 11,757 tuples. It contains the criminal history, jail and prison time, demographics and Compas risk scores for defendants from Broward County from 2013 and 2014. The data is usually used to predict whether a person will re-offend within the next 2 years.

- **Healthcare** ⁷: A dataset containing 11 attributes and 1000 tuples. It contains personal data of patients, such as the income, number of complications, county, race, age.

All datasets were preprocessed so that tuples with missing values in the relevant columns were dropped, negative numerical values were replaced by zeros since their domains are nonnegative, categorical values were encoded. For the IPUMS-CPS dataset, the attribute AGE was discretized as 10 years per range, e.g., $[0, 10]$ is considered a single value. Also, we only included tuples with attribute INCTOT (the total income) smaller than 200K to avoid outliers, as in [77].

Fairness criteria. In Section 2.2, we formalized our fairness criteria as conditional independence statements. Here, we instantiated these generic measures for each dataset by choosing concrete protected, response, and admissible attributes; the resulting criteria are listed in Table 2. For every dataset, we numbered the corresponding fairness criteria 1 to 4 and used this numbering in the figures. The four criteria were chosen to reflect natural desiderata for protected attributes (e.g., sex or race) to be independent from the attribute that will be predicted (e.g., income).

⁴<https://cps.ipums.org/cps>

⁵<https://kaggle.com/datasets/berkayalan/stack-overflow-annual-developer-survey-2024>

⁶<https://github.com/propublica/compas-analysis/blob/master/compas-scores.csv>

⁷https://github.com/stefan-grafberger/mlinspect/tree/master/example_pipelines

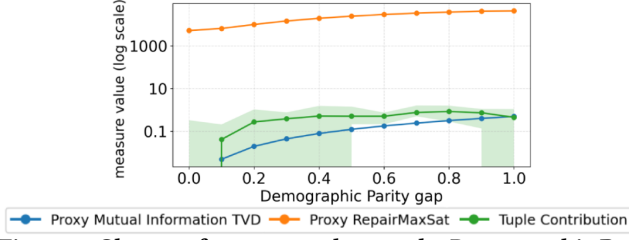


Figure 4: Change of measure values as the Demographic Parity in the dataset gradually increases.

Algorithm variations and optimizations. Since there are no existing baselines for unfairness measures, we compared Algorithms 1 to 3 to several variations and employed a heuristic for Algorithm 2:

- **Non-private versions:** Algorithms 1 to 3 were compared against their non-private counterparts, i.e., setting $\epsilon = \infty$. We additionally ran a dedicated experiment in which the privacy budget varies from 0.1 to 10. For the rest of the experiments, we use a privacy budget of $\epsilon = 1$, unless otherwise specified.
- **Varying k in Algorithm 3:** Algorithm 3 was examined with different values of k for $\mathcal{U}_{TC}$, from 100 to 30K. In all other experiments, we fixed $k = 500$.
- **Heuristic for Algorithm 2:** Due to scalability issues of the SAT solver in line 4 of Algorithm 2, we employed a heuristic that partitions the database to consecutive chunks of 100 tuples, then runs Algorithm 2 separately for each chunk and each criterion, and sums up the results to yield an estimate of the $\mathcal{U}_R^{SAT}$ value. Assuming that the tuple order is preserved in neighboring databases, as is standard for bounded DP [68], the sensitivity of this heuristic remains identical to $\Delta_{\mathcal{U}_R^{SAT}}$, as proven in Proposition A.17.

Relative $L1$ error. In some of the experiments we measured relative $L1$ error of the algorithms. For a given ϵ , we define the relative error as $\frac{|X-Y|}{\max(Y, e^{-100})}$, where X is the output of the algorithm with privacy budget ϵ , and Y is the output of the algorithm without privacy. The use of e^{-100} in the denominator is meant to avoid division by 0.

5.2 Unfairness Detection

We studied how the proposed unfairness measures react to gradually increasing unfairness in a dataset. We constructed a binary synthetic dataset with attributes sex and income > 50K containing 100,000 tuples, with an equal number of male and female records. Initially, the dataset was completely fair. We then progressively flipped an increasing fraction of male records to have income above 50K and female records to have income below 50K, thereby monotonically increasing the demographic parity gap $|Pr_D(\text{income} > 50K \mid \text{sex} = M) - Pr_D(\text{income} > 50K \mid \text{sex} = F)|$. In Figure 4, we plot the values of the measures as a function of the demographic parity gap, with privacy budget $\epsilon = 10$. We observe that $\mathcal{U}_R^{SAT}$ (as computed by Algorithm 2 with our heuristic) exhibits almost linear growth as the demographic parity gap increases, while $\mathcal{U}_{MI}^{TVD}$ (as computed by Algorithm 1) exhibits growth that appears logarithmic. $\mathcal{U}_{TC}$ (as computed by Algorithm 3) shows a slight and noisy increase in values, which may suggest that $k = 500$ inhibits its ability to detect the gap in this dataset. It also decreases slightly toward the end of the graph, since when the dataset becomes fully polarized, changing one tuple has no effect on unfairness, and thus

the MD of every tuple decreases. Overall, all measures correctly detect small changes in unfairness in the dataset.

5.3 Scalability

Figure 5 depicts the effect of data size and the number of fairness criteria on the runtime of the algorithms. In Figure 5a, we plot the runtime of the algorithms for each dataset as a function of the number of tuples. Across all datasets, the computation of Algorithm 2 is consistently the slowest and even exceeds the time limit for IPUMS-CPS at more than 600,000 tuples. Algorithm 3 is frequently the fastest measure to compute, with an exception being the IPUMS-CPS dataset, on which the runtime is higher and mostly insensitive to the number of tuples. This is attributed to the fact that, as formulated in Section 4.3, Algorithm 3 uses a contingency table over the attribute sets, whose cardinalities are higher for IPUMS-CPS than for the other datasets. Finally, Algorithm 1 is on average faster than Algorithm 2 but slower than Algorithm 3. We can see that the runtimes of all the algorithms increase steadily with the number of tuples. Still, runtimes for smaller numbers of tuples are sometimes higher and more volatile because they are more sensitive to randomness in the sampled data.

In Figure 5b, we show the runtime of the algorithms for each dataset as a function of the number of fairness criteria. For each dataset, we begin with only the first criterion in Table 2 and then progressively add more, such that the final measurement includes all criteria for that dataset. We can see that, for all the algorithms, increasing the number of criteria inflates the runtime more gradually than increasing the number of tuples. Overall, Algorithm 3 is the least affected by increasing the number of tuples and the number of criteria for moderately dense datasets, and it is the fastest algorithm in almost all settings. However, its runtime increases noticeably for sparse datasets such as IPUMS-CPS. As for the other algorithms, Algorithm 2 is generally the slowest and most affected algorithm, while Algorithm 1 lies in between.

5.4 Accuracy and Privacy Tradeoff

Figure 6 shows the effect of increasing the privacy budget on the relative $L1$ error of the algorithms. Algorithm 3 is consistently the least accurate of the three algorithms, due to its sensitivity being directly proportional to k and inversely proportional to the minimum group size per admissible value (Proposition 3.12). Consequently, for $k = 500$, the noise added for all datasets is larger than that added for the other algorithms; for datasets with small minimum group sizes, such as IPUMS-CPS and Adult, this noise is particularly high. The parameter k can be calibrated for each dataset to reduce the noise, as shown in a later experiment, thereby reducing the relative $L1$ error. Algorithm 2 is the most accurate of the three algorithms. This is because its values are much larger in magnitude, so the added DP noise changes the true value only by a small fraction, leading to a smaller relative error. Additionally, the error of Algorithm 2 is the most stable across all datasets, since its sensitivity does not depend on the number of tuples (Proposition 3.10), causing the error to be primarily determined by ϵ . Algorithm 1 is less noisy than Algorithm 3, which is consistent with its low sensitivity (Proposition 3.4), but more noisy than Algorithm 2. Moreover, Algorithm 1 exhibits large variance between runs, since its values are small and

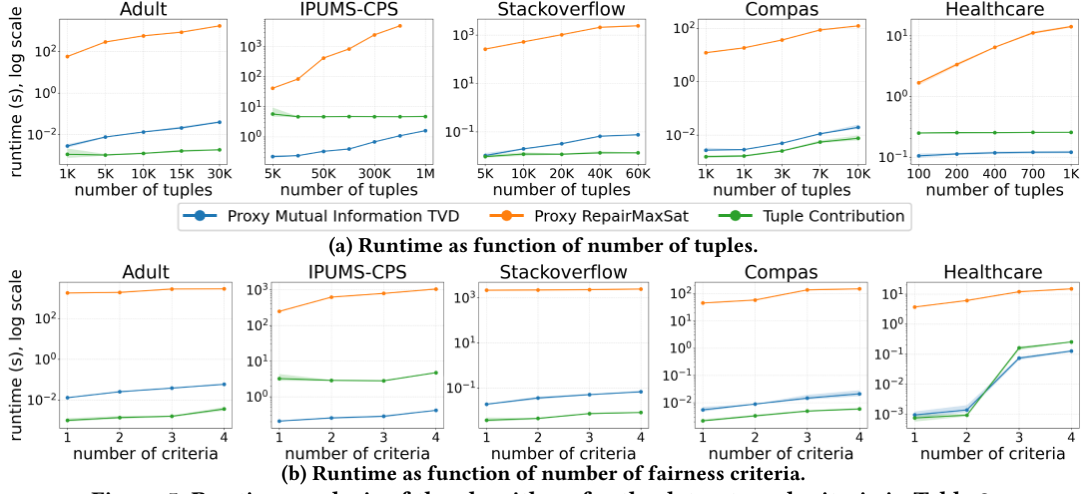


Figure 5: Runtime analysis of the algorithms for the datasets and criteria in Table 2.

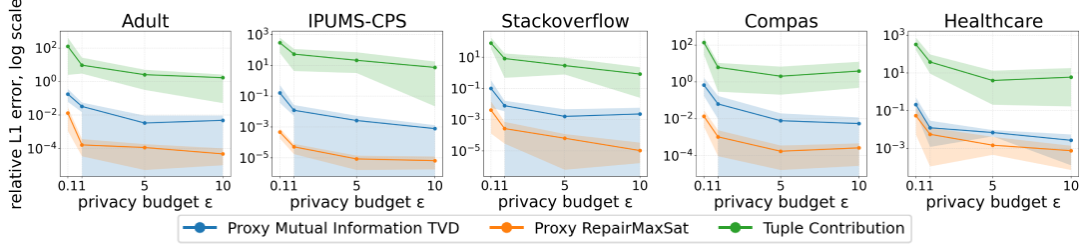


Figure 6: Relative L1 error as function of privacy budget for the datasets and criteria from Table 2.

therefore easily distorted by the added noise. However, for small values of ϵ , the noise dominates the output, leading to reduced variance in the error. Overall, the relative errors of all algorithms steadily decrease as the privacy budget increases, as expected, with the sharpest declining segment occurring up to $\epsilon = 1$.

5.5 Measure Drill-Down

Proxy faithfulness of $\mathcal{U}_{MI}^{TVD}$. Following Figure 3, Figure 7 shows a comparison of $\mathcal{U}_{MI}$ and $\mathcal{U}_{MI}^{TVD}$ values for every criterion from Table 2, with privacy budget $\epsilon = 1$. For every dataset, $\mathcal{U}_{MI}^{TVD}$ closely tracks $\mathcal{U}_{MI}$, with average Kendall’s tau correlation [46] being 92%. Specifically, an increase in $\mathcal{U}_{MI}$ corresponds to an increase in $\mathcal{U}_{MI}^{TVD}$, and vice versa, as we would expect from a good proxy. However, this proportionality does not extend across datasets, since for the Healthcare and Compas datasets the same value of $\mathcal{U}_{MI}$ (0.09) corresponds to different values of $\mathcal{U}_{MI}^{TVD}$. Within each dataset, we can further glean the level of unfairness for a criterion by comparing the values of the same measure for it and for another criterion. For example, for IPUMS-CPS, both measures assign higher values to criterion 4 (HEALTH $\perp$ INCTOT | AGE) than to criterion 3 (HEALTH $\perp$ MARST | AGE), meaning that the predicted unfairness for criterion 4 is larger. Intuitively, given the age of a person, their income depends on health more than their marital status does.

Effect of k on $\mathcal{U}_{TC}$. Figure 8 depicts the effect of the parameter k on the value and relative L1 error of Algorithm 3 (computing $\mathcal{U}_{TC}$). In Figure 8a, we plot the value of Algorithm 3 as k increases. Overall, as expected, the value is monotone non-decreasing in k because, as

k grows, we add an increasing number of non-negative $\mathcal{MD}$ values to the resulting sum. Almost all datasets reach a ‘plateau’ starting from some value of k , meaning that a subset of tuples carries most of the $\mathcal{MD}$ of the dataset, while the remaining tuples contribute only marginally as k increases. For example, in Healthcare, most tuples have large $\mathcal{MD}$ values, and thus $\mathcal{U}_{TC}$ continues to grow almost linearly in k . In contrast, for IPUMS-CPS, many tuples have small $\mathcal{MD}$ values, so the trend does not plateau and remains noisy.

Figure 8b depicts the relative L1 error as a function of k . For this experiment, we increased the number of repetitions for each data point to 50 in order to observe the trends more clearly. Although the mean relative error remains noisy, it overall increases with k . This behavior arises because the sensitivity of $\mathcal{U}_{TC}$, and consequently the magnitude of the added DP noise, is directly proportional to k (Proposition 3.12). Furthermore, the error is larger and exhibits greater variability for datasets with many distinct values, such as IPUMS-CPS. We recognize that selecting an optimal value of k for $\mathcal{U}_{TC}$ is a challenging problem and leave this for future work.

Chunk heuristic effect on $\mathcal{U}_R^{SAT}$. We compared the versions of Algorithm 2 with and without the heuristic (i.e., computing $\mathcal{U}_R^{SAT}$ with chunks of size 100, and original $\mathcal{U}_R^{SAT}$), with privacy budget $\epsilon = 1$. The computation of Algorithm 2 with the heuristic is noticeably faster than when computing $\mathcal{U}_R^{SAT}$ directly, reducing the total runtime by 56.26% across all datasets. The values for the heuristic mirror the trend of the values for the original $\mathcal{U}_R^{SAT}$ as we show in Figure 10. Across all datasets, the Kendall’s tau correlation [46] is 72% due to slightly different results for criteria 1,2 of IPUMS-CPS and criterion 3 for Stackoverflow (Table 2). We

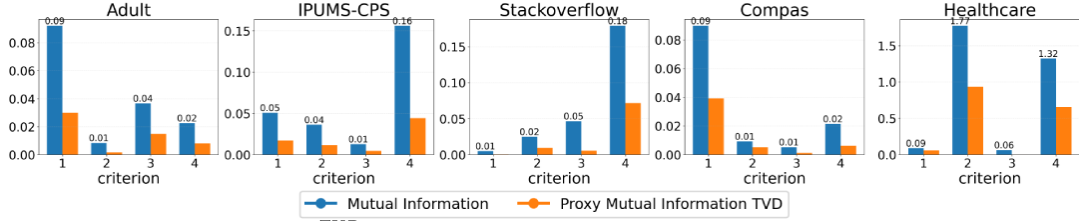


Figure 7: Faithfulness of noisy $\mathcal{U}_{MI}^{TVD}$ to noisy $\mathcal{U}_{MI}$ over different datasets and fairness criteria from Table 2.

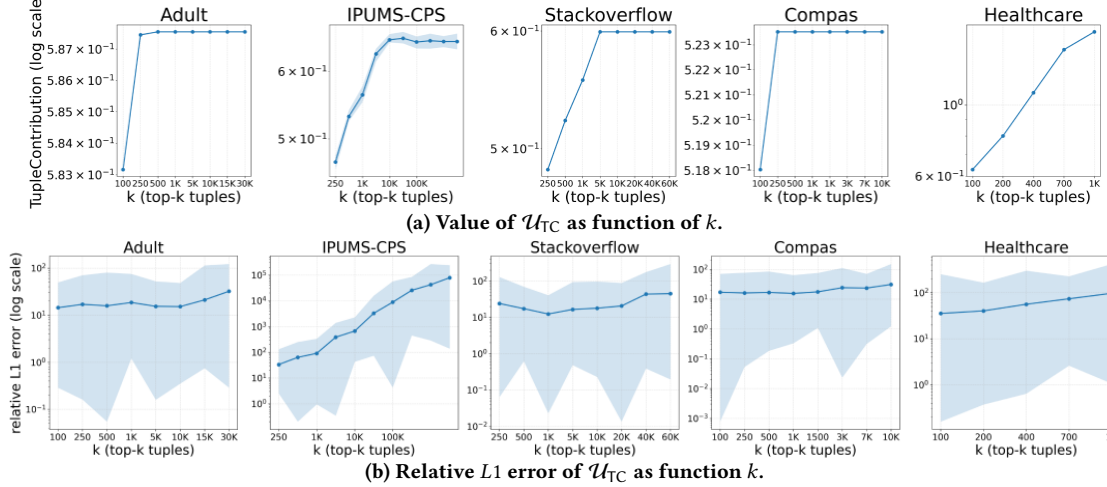


Figure 8: Effect of k on $\mathcal{U}_{TC}$ in terms of value and relative $L1$ error.

ignored criterion 4 for Stackoverflow in the calculations because Algorithm 2 without the heuristic exceeded the time limit. Without these criteria, the Kendall’s tau correlation is 100%.

6 RELATED WORK

Research on fairness in data management intersects several areas, including data valuation, algorithmic fairness, private data pricing, and inconsistency assessment in relational systems. Although these directions provide important foundations, *none develops dedicated measures for data centric fairness tailored for use in the differential privacy setting*, which is the focus of this paper.

A central line of work studies Data Valuation Assessment (DVA) in machine learning, where the goal is to estimate the contribution of individual records or subsets of data to predictive performance [5, 10, 47]. These methods quantify the marginal value of data with respect to accuracy or loss, and emphasize model driven utility rather than fairness or structural properties of the data itself.

Private data valuation methods provide mechanisms for pricing or compensating for the use of sensitive data without addressing fairness concerns [34, 53, 85]. These works typically quantify sensitivity, privacy loss, or economic value but do not analyze disparate impact or structural imbalances in the data.

Inconsistency measures for relational data and integrity constraints [58, 61, 62, 70] offer tools for quantifying violations of integrity constraints. These measures are inspired by long standing work in Knowledge Representation and Logic [36, 37, 43, 49, 50, 67, 79], which develops principles for reasoning under inconsistency. While conceptually related, these methods address logical coherence rather than fairness across protected groups.

Finally, fairness driven data repair methods aim to modify datasets to remove discriminatory patterns as pre or post-processing solutions [4, 12, 29, 39, 45, 74]. These interventions assume some fairness criterion but do not provide measures designed specifically for assessing fairness within the DP context.

7 CONCLUSION AND LIMITATIONS

We presented a principled framework for quantifying data unfairness under differential privacy, bridging concepts from probabilistic dependence, data repair, and contribution analysis. Our measures satisfy desirable consistency properties, exhibit low sensitivity, and remain tractable under standard DP mechanisms. Our three measures collectively provide complementary perspectives on data level bias. Empirical results across multiple datasets confirm that our measures preserve fidelity to non private baselines despite privacy noise. Overall, the framework lays a foundation for systematic evaluation of fairness in privacy protected data.

While our framework provides a principled foundation for measuring database unfairness under differential privacy, several limitations remain. First, the $\mathcal{U}_R^{SAT}$ relies on a MaxSAT-based proxy whose practical instantiation uses block-wise heuristics. This improves scalability but weakens the estimation ability associated with the original formulation. Second, a principled approach for choosing k would benefit users and make $\mathcal{U}_{TC}$ more user-friendly. Finally, our measures operate purely at the associational level and do not account for causal structure, data collection bias, or label noise, which may also influence fairness assessments.

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A THEOREMS AND PROOFS

A.1 Proofs for Section 3

PROOF OF PROPOSITION 3.3. According to the Pinsker's inequality [19]:

$$\text{TVD}(Pr_D(P, Y), Pr_D(P)Pr_D(Y))^2 \leq \frac{1}{2} D_{\text{KL}}(Pr_D(P, Y) \parallel Pr_D(P)Pr_D(Y)) = \frac{1}{2} \mathcal{U}_{\text{MI}}(P, Y)$$

it implies that

$$\mathcal{U}_{\text{MI}}^{TVD}(P \perp\!\!\!\perp Y, D) = 2 \cdot \text{TVD}(Pr_D(P, Y), Pr_D(P)Pr_D(Y))^2 \leq \mathcal{U}_{\text{MI}}(P, Y)$$

From the other side, according to the reverse Pinsker's inequality, for any two distributions P and Q such that $Q(z) > 0$ for all z it holds that

$$\text{TVD}(P, Q)^2 \geq \frac{1}{2} \cdot \min_z Q(z) \cdot D_{\text{KL}}(P \parallel Q)$$

Setting $P = Pr_D(P, Y)$ and $Q = Pr_D(P)Pr_D(Y)$, it implies that

$$\text{TVD}(Pr_D(P, Y), Pr_D(P)Pr_D(Y))^2 \geq \frac{1}{2} \cdot \alpha \cdot \mathcal{U}_{\text{MI}}(P, Y)$$

where $\alpha := \min_{(p, y)} Pr_D(P = p)Pr_D(Y = y)$ over the support of $Pr_D(P, Y)$.

Multiplying both sides by 2, we get

$$\mathcal{U}_{\text{MI}}^{TVD}(P \perp\!\!\!\perp Y, D) = 2 \cdot \text{TVD}(Pr_D(P, Y), Pr_D(P)Pr_D(Y))^2 \geq \alpha \cdot \mathcal{U}_{\text{MI}}(P, Y)$$

Therefore

$$\alpha \cdot \mathcal{U}_{\text{MI}}(P, Y) \leq \mathcal{U}_{\text{MI}}^{TVD}(P \perp\!\!\!\perp Y, D) \leq \mathcal{U}_{\text{MI}}(P, Y)$$

where $\alpha := \min_{(p, y)} Pr_D(P = p)Pr_D(Y = y)$.

Then, if we define $X = Pr_D(P, Y)$, $Y = Pr_D(P)Pr_D(Y)$, we would get that the following holds

$$\alpha \cdot \mathcal{U}_{\text{MI}}(P, Y) \mathcal{U}_{\text{MI}}^{TVD}(P \perp\!\!\!\perp Y, D) \leq \mathcal{U}_{\text{MI}}(P, Y)$$

where $\alpha = \min_{z \in Z, Y(z) > 0} Y(z)$. $\square$

PROOF OF PROPOSITION 3.4 PARTS 1 AND 2. **Positivity:**

By the definition of total variation distance, for any two probability distributions P and Q it holds that

$$\text{TVD}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)| \geq 0$$

and it holds that $\text{TVD}(P, Q) = 0$ iff $P = Q$ by the properties of the absolute value.

For any database D and any fairness criterion $F = P \perp\!\!\!\perp Y \mid A \in \mathcal{F}$ where A can be absent, it holds that

$$\begin{aligned} \mathcal{U}_{\text{MI}}^{TVD}(P \perp\!\!\!\perp Y \mid A, D) &= 2 \cdot (\text{TVD}(Pr_D(P, Y \mid A), Pr_D(P \mid A)Pr_D(Y \mid A)))^2 \\ &\geq \text{TVD}(Pr_D(P, Y \mid A), Pr_D(P \mid A)Pr_D(Y \mid A)) \\ &\geq 0 \end{aligned}$$

and

$$\begin{aligned} \mathcal{U}_{\text{MI}}^{TVD}(P \perp\!\!\!\perp Y \mid A, D) &= 0 \\ \iff 2 \cdot (\text{TVD}(Pr_D(P, Y \mid A), Pr_D(P \mid A)Pr_D(Y \mid A)))^2 &= 0 \\ \iff \text{TVD}(Pr_D(P, Y \mid A), Pr_D(P \mid A)Pr_D(Y \mid A)) &= 0 \\ \iff Pr_D(P, Y \mid A) = Pr_D(P \mid A)Pr_D(Y \mid A) & \\ \iff F(D) = 1 \end{aligned}$$

We can notice that the above also holds for any a set of fairness criteria $\mathcal{F}$ since $\mathcal{U}_{\text{MI}}^{TVD}(\mathcal{F}, D) := \sum_{F \in \mathcal{F}} \mathcal{U}_{\text{MI}}^{TVD}(F, D)$.

Monotonicity:

For any database D and fairness criteria $\mathcal{F}, \mathcal{F}'$ such that $\mathcal{F} \subseteq \mathcal{F}'$, it holds that

$$\mathcal{U}_{\text{MI}}^{TVD}(\mathcal{F}, D) = \sum_{F \in \mathcal{F}} \mathcal{U}_{\text{MI}}^{TVD}(F, D) \leq \sum_{F' \in \mathcal{F}'} \mathcal{U}_{\text{MI}}^{TVD}(F', D) = \mathcal{U}_{\text{MI}}^{TVD}(\mathcal{F}', D)$$

This follows from the fact that $\mathcal{U}_{\text{MI}}^{TVD}$ is non-negative (from the Positivity property), and so its sum is monotonic. $\square$

We will first state and prove the range and the sensitivity bounds of $\mathcal{U}_{\text{MI}}^{TVD}$ for a single fairness criterion, and then we will use them to prove the range and the sensitivity bounds for a set of criteria (Parts 3 and 4 of Proposition 3.4).

LEMMA A.1 ($\mathcal{U}_{\text{MI}}^{TVD}$ IS SUITABLE FOR DP FOR A SINGLE FAIRNESS CRITERION). *Given a database D of size n , the following holds:*

- (1) *The range of $\mathcal{U}_{\text{MI}}^{TVD}$ is $[0, 2]$ for a single criterion F .*
- (2) *The sensitivity of $\mathcal{U}_{\text{MI}}^{TVD}$ for an unconditional fairness criterion $F = P \perp\!\!\!\perp Y$ is at most $\frac{12}{n}$ and for a conditional fairness criterion, $F = P \perp\!\!\!\perp Y \mid A$, is at most $\frac{16}{n}$.*

PROOF OF LEMMA A.1 PART 1. For any P and Q two probability distributions, it holds that:

$$0 \leq \frac{1}{2} \sum_x |P(x) - Q(x)| \leq \frac{1}{2} (|1 - 0| + |0 - 1|) = 1$$

by the properties of probability. The upper bound is for the case when all probability mass in P is at one outcome, while all probability mass in Q is at the other. In this case, only two terms contribute to the sum, and both contribute 1.

Therefore by the definition of TVD it holds that:

$$0 = 2 \cdot 0^2 \leq 2 \cdot (\text{TVD}(P, Q))^2 \leq 2 \cdot 1^2 = 2$$

And so, by the definition, the range of $\mathcal{U}_{\text{MI}}^{TVD}$ is $[0, 2]$. $\square$

Now we will state and prove a few lemmas that we will use in a later proof of Lemma A.1 Part 2.

LEMMA A.2 (REVERSE TRIANGLE INEQUALITY). *Given $x, y \in \mathbb{R}$, the following holds*

$$||x| - |y|| \leq |x - y|$$

PROOF. From the triangle inequality,

$$|x| = |(x - y) + y| \leq |x - y| + |y|$$

Rearranging, we get

$$|x| - |y| \leq |x - y|$$

Similarly, by symmetry,

$$|y| = |(y - x) + x| \leq |y - x| + |x| = |x - y| + |x|$$

Rearranging, we get

$$|y| - |x| \leq |x - y|$$

and multiplying both sides by -1 we get

$$-|x - y| \leq -(|y| - |x|) = |x| - |y|$$

Combining the two, we get

$$-|x - y| \leq |x| - |y| \leq |x - y|$$

which by the definition of the absolute value means

$$||x| - |y|| \leq |x - y|$$

$\square$

LEMMA A.3 (SENSITIVITY OF THE SQUARE). *Let $f : \mathcal{D} \rightarrow \mathbb{R}$ be a function with sensitivity Δ_f . Define $g(D) := f(D)^2$. Then the sensitivity of g satisfies:*

$$\Delta_g := \max_{D \sim D'} |g(D) - g(D')| \leq 2 \cdot \Delta_f \cdot \max(\text{Dom}(f))$$

PROOF. Let $D \sim D'$ be neighboring databases. Define $g(D) = f(D)^2$, so

$$\begin{aligned} |g(D) - g(D')| &= |f(D)^2 - f(D')^2| \\ &= |(f(D) - f(D')) \cdot (f(D) + f(D'))| \\ &\leq |f(D) - f(D')| \cdot |f(D) + f(D')| \\ &\leq |f(D) - f(D')| \cdot 2 \cdot \max(\text{Dom}(f)) \\ &\leq \Delta_f \cdot 2 \cdot \max(\text{Dom}(f)) \end{aligned}$$

Taking the maximum over all neighboring databases, we get

$$\Delta_g \leq 2 \cdot \Delta_f \cdot \max(\text{Dom}(f))$$

□

PROOF OF LEMMA A.1 PART 2. **Unconditional case:**

Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of $t'' = (p_{t''}, y_{t''})$ and one less occurrence of $t' = (p_{t'}, y_{t'})$ than D .

From Definition 3.2, by the triangle inequality and Lemma A.2, we have the following

$$|\text{TVD}(Pr_{D'}(P, Y), Pr_{D'}(P)Pr_{D'}(Y)) - \text{TVD}(Pr_D(P, Y), Pr_D(P)Pr_D(Y))| \quad (1)$$

$$\begin{aligned} &= \left| \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_{D'}(P = p, Y = y) - Pr_{D'}(P = p)Pr_{D'}(Y = y)| \right. \\ &\quad \left. - \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_D(P = p, Y = y) - Pr_D(P = p)Pr_D(Y = y)| \right| \quad (2) \end{aligned}$$

$$\leq \left| \frac{1}{2} \sum_{p \in P, y \in Y} (Pr_{D'}(P = p, Y = y) - Pr_{D'}(P = p)Pr_{D'}(Y = y) - Pr_D(P = p, Y = y) + Pr_D(P = p)Pr_D(Y = y)) \right| \quad (3)$$

$$\begin{aligned} &= \frac{1}{2} \left| \sum_{p \in P, y \in Y} (Pr_{D'}(P = p, Y = y) - Pr_D(P = p, Y = y)) \right. \\ &\quad \left. + \sum_{p \in P, y \in Y} (Pr_D(P = p)Pr_D(Y = y) - Pr_{D'}(P = p)Pr_{D'}(Y = y)) \right| \quad (4) \end{aligned}$$

$$\begin{aligned} &\leq \frac{1}{2} \left| \sum_{p \in P, y \in Y} Pr_{D'}(P = p, Y = y) - Pr_D(P = p, Y = y) \right| \\ &\quad + \frac{1}{2} \left| \sum_{p \in P, y \in Y} Pr_{D'}(P = p)Pr_{D'}(Y = y) - Pr_D(P = p)Pr_D(Y = y) \right| \quad (5) \end{aligned}$$

$$\begin{aligned} &\leq \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_{D'}(P = p, Y = y) - Pr_D(P = p, Y = y)| \\ &\quad + \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_{D'}(P = p)Pr_{D'}(Y = y) - Pr_D(P = p)Pr_D(Y = y)| \quad (6) \end{aligned}$$

The transition from (2) to (3) is due to Lemma A.2, the transitions from (4) to (5) and from (5) to (6) are due to triangle inequality.

Looking at the first part of the sum separately, we get

$$\begin{aligned} &\frac{1}{2} \sum_{p \in P, y \in Y} |Pr_{D'}(P = p, Y = y) - Pr_D(P = p, Y = y)| \\ &= \frac{1}{2} (|Pr_{D'}(P = p_{t'}, Y = y_{t'}) - Pr_D(P = p_{t'}, Y = y_{t'})| \\ &\quad + |Pr_{D'}(P = p_{t''}, Y = y_{t''}) - Pr_D(P = p_{t''}, Y = y_{t''})|) \\ &\leq \frac{1}{2} \cdot \left(\frac{1}{n} + \frac{1}{n} \right) \\ &= \frac{1}{n} \end{aligned}$$

Looking at the second part of the sum, we get

$$\frac{1}{2} \sum_{p \in P, y \in Y} |Pr_{D'}(P = p)Pr_{D'}(Y = y) - Pr_D(P = p)Pr_D(Y = y)| \quad (7)$$

$$\begin{aligned} &= \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_{D'}(P = p)Pr_{D'}(Y = y) + Pr_D(P = p)Pr_{D'}(Y = y) \\ &\quad - Pr_D(P = p)Pr_{D'}(Y = y) - Pr_D(P = p)Pr_D(Y = y)| \quad (8) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \sum_{p \in P, y \in Y} |(Pr_{D'}(P = p) - Pr_D(P = p))Pr_{D'}(Y = y) \\ &\quad + Pr_D(P = p)(Pr_{D'}(Y = y) - Pr_D(Y = y))| \quad (9) \end{aligned}$$

$$\begin{aligned} &\leq \frac{1}{2} \sum_{p \in P, y \in Y} (|Pr_{D'}(P = p) - Pr_D(P = p)|Pr_{D'}(Y = y) \\ &\quad + Pr_D(P = p)|Pr_{D'}(Y = y) - Pr_D(Y = y)|) \quad (10) \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \left(\sum_{p \in P} |Pr_{D'}(P = p) - Pr_D(P = p)| \sum_{y \in Y} Pr_{D'}(Y = y) \right. \\ &\quad \left. + \sum_{y \in Y} |Pr_{D'}(Y = y) - Pr_D(Y = y)| \sum_{p \in P} Pr_D(P = p) \right) \quad (11) \end{aligned}$$

$$= \frac{1}{2} \left(\sum_{p \in P} |Pr_{D'}(P = p) - Pr_D(P = p)| \cdot 1 + \sum_{y \in Y} |Pr_{D'}(Y = y) - Pr_D(Y = y)| \cdot 1 \right) \quad (12)$$

$$= \frac{1}{2} \sum_{p \in P} |Pr_{D'}(P = p) - Pr_D(P = p)| + \frac{1}{2} \sum_{y \in Y} |Pr_{D'}(Y = y) - Pr_D(Y = y)| \quad (13)$$

where the transition from (9) to (10) is due to applying the triangle inequality, the transition from (11) to (12) is due to the fact that the sum of probabilities for all the values is 1.

In particular,

$$\begin{aligned} &\frac{1}{2} \sum_{p \in P} |Pr_{D'}(P = p) - Pr_D(P = p)| \\ &\leq \frac{1}{2} \cdot \left(\frac{1}{n} + \frac{1}{n} \right) \\ &= \frac{1}{n} \end{aligned}$$

A similar expression can be achieved by swapping P and p for Y and y , respectively.

Putting it all together,

$$\begin{aligned} &|\text{TVD}(Pr_{D'}(P, Y), Pr_{D'}(P)Pr_{D'}(Y)) - \text{TVD}(Pr_D(P, Y), Pr_D(P)Pr_D(Y))| \\ &\leq \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_{D'}(p, y) - Pr_D(p, y)| \\ &\quad + \frac{1}{2} \sum_{p \in P, y \in Y} |Pr_{D'}(p)Pr_{D'}(y) - Pr_D(p)Pr_D(y)| \\ &\leq \frac{1}{n} + \frac{1}{2} \sum_{p \in P} |Pr_{D'}(p) - Pr_D(p)| + \frac{1}{2} \sum_{y \in Y} |Pr_{D'}(y) - Pr_D(y)| \\ &\leq \frac{3}{n} \end{aligned}$$

Taking the maximum over all neighboring databases $D \sim D'$, we get

$$\Delta_{\text{TVD}} \leq \frac{3}{n}$$

where $n = |D|$.

From Lemma A.3, it holds that

$$\begin{aligned} \Delta_{\mathcal{U}_{\text{MI}}^{\text{TVD}}} &= \Delta_{2\text{TVD}^2} \\ &= 2 \cdot \Delta_{\text{TVD}^2} \\ &\leq 2 \cdot 2 \cdot \Delta_{\text{TVD}} \cdot \max(\text{Dom}(\text{TVD})) \\ &\stackrel{\text{TVD} \leq 1}{\leq} 4 \cdot \frac{3}{n} \cdot 1 \\ &= \frac{12}{n} \end{aligned}$$

Therefore, the sensitivity bound for the unconditional case is $\frac{12}{n}$.

Conditional case:

Similarly, assume that D' contains one more occurrence of $t'' = (p_{t''}, y_{t''}, a_{t''})$ and one less occurrence of $t' = (p_{t'}, y_{t'}, a_{t'})$ than D .

Denote $f_D(a) := \mathcal{U}_{\text{MI}}^{TVD}(P \perp Y \mid A = a, D)$, and similarly $f_{D'}(a)$. Then from Definition 3.2 it holds that

$$\left| \mathcal{U}_{\text{MI}}^{TVD}(P \perp Y \mid A, D') - \mathcal{U}_{\text{MI}}^{TVD}(P \perp Y \mid A, D) \right| \quad (14)$$

$$= \left| \sum_a (Pr_{D'}(A = a) f_{D'}(a) - Pr_D(A = a) f_D(a)) \right| \quad (15)$$

$$\leq \sum_a |Pr_{D'}(A = a) f_{D'}(a) - Pr_D(A = a) f_D(a)| \quad (16)$$

$$= \sum_a \left| \left((Pr_{D'}(A = a) - Pr_D(A = a)) f_{D'}(a) \right) + \left(Pr_D(A = a) (f_{D'}(a) - f_D(a)) \right) \right| \quad (17)$$

$$\leq \sum_a \left| \left((Pr_{D'}(A = a) - Pr_D(A = a)) f_{D'}(a) \right) \right| + \left| Pr_D(A = a) (f_{D'}(a) - f_D(a)) \right| \quad (18)$$

$$\leq \sum_a |Pr_{D'}(A = a) - Pr_D(A = a)| |f_{D'}(a)| + \sum_a Pr_D(A = a) |f_{D'}(a) - f_D(a)| \quad (19)$$

where the transition from (15) to (17), and from (17) to (18) is due to the triangle inequality; and the transition from (18) to (19) is due to the fact that $xy \leq |x||y|$ for $x, y \in \mathbb{R}$.

Looking at the first sum separately, because only $a \in \{a_{t'}, a_{t''}\}$ change, and since $|f_{D'}(a)| \leq 2$ by the Range property from Proposition 3.4 proved earlier, then

$$\sum_a |Pr_{D'}(A = a) - Pr_D(A = a)| |f_{D'}(a)| \leq \left(\frac{1}{n} + \frac{1}{n} \right) \cdot 2 = \frac{4}{n}$$

Looking at the second sum, by the unconditional case and by the Lemma A.3, for a fixed a with $n_a = |\{t \in D \mid t[A] = a\}|$, it holds that the sensitivity of $2TVD^2$ is at most $\frac{12}{n_a}$ when both removing t' and adding t'' are included in the computation. So $|f_{D'}(a_{t'}) - f_D(a_{t'})| = \frac{1}{2} \cdot \frac{12}{n_{a_{t'}}}$, and similarly for t'' . Therefore

$$\begin{aligned} & \sum_a Pr_D(A = a) |f_{D'}(a) - f_D(a)| \\ &= Pr_D(A = a_{t'}) |f_{D'}(a_{t'}) - f_D(a_{t'})| + Pr_D(A = a_{t''}) |f_{D'}(a_{t''}) - f_D(a_{t''})| \\ &\leq \frac{n_{a_{t'}}}{n} \cdot \frac{6}{n_{a_{t'}}} + \frac{n_{a_{t''}}}{n} \cdot \frac{6}{n_{a_{t''}}} \\ &= \frac{12}{n} \end{aligned}$$

Putting it all together,

$$\Delta_{\mathcal{U}_{\text{MI}}^{TVD}} = \max_{D' \sim D} \left| \mathcal{U}_{\text{MI}}^{TVD}(P \perp Y \mid A, D') - \mathcal{U}_{\text{MI}}^{TVD}(P \perp Y \mid A, D) \right| \leq \frac{4}{n} + \frac{12}{n} = \frac{16}{n}$$

Therefore, the sensitivity bound for the conditional case is $\frac{16}{n}$. $\square$

PROOF OF PROPOSITION 3.4 PARTS 3 AND 4. Range:

By Definition 3.2 and by Lemma A.1, it holds that

$$\mathcal{U}_{\text{MI}}^{TVD}(\mathcal{F}, D) = \sum_{F \in \mathcal{F}} \mathcal{U}_{\text{MI}}^{TVD}(F, D) \geq \sum_{F \in \mathcal{F}} 0 = 0$$

and

$$\mathcal{U}_{\text{MI}}^{TVD}(\mathcal{F}, D) = \sum_{F \in \mathcal{F}} \mathcal{U}_{\text{MI}}^{TVD}(F, D) \leq \sum_{F \in \mathcal{F}} 2 = 2|\mathcal{F}|$$

Sensitivity:

From Lemma A.1, the sensitivity for a single fairness criterion F is at most

$$\Delta_{\mathcal{U}_{\text{MI}}^{TVD}(F, D)} \leq \frac{16}{n}$$

Therefore, for $D' \sim D$, using the triangle inequality and linearity of the sum, it holds that

$$\begin{aligned} |\mathcal{U}_{\text{MI}}^{TVD}(\mathcal{F}, D') - \mathcal{U}_{\text{MI}}^{TVD}(\mathcal{F}, D)| &= \left| \sum_{F \in \mathcal{F}} (\mathcal{U}_{\text{MI}}^{TVD}(F, D') - \mathcal{U}_{\text{MI}}^{TVD}(F, D)) \right| \\ &\leq \sum_{F \in \mathcal{F}} |\mathcal{U}_{\text{MI}}^{TVD}(F, D') - \mathcal{U}_{\text{MI}}^{TVD}(F, D)| \\ &\leq \sum_{F \in \mathcal{F}} \Delta_{\mathcal{U}_{\text{MI}}^{TVD}(F, D)} \\ &\leq \frac{16|\mathcal{F}|}{n} \end{aligned}$$

PROPOSITION A.4 ($\mathcal{U}_R$ SATISFIES THE DESIRED PROPERTIES). $\square$

- (1) $\mathcal{U}_R$ satisfies the Positivity property.
- (2) $\mathcal{U}_R$ satisfies the Monotonicity property.
- (3) The range of $\mathcal{U}_R$ is $[0, n]$ for a set of criteria $\mathcal{F}$.
- (4) The sensitivity of $\mathcal{U}_R$ is 1 for a set of criteria $\mathcal{F}$.

PROOF OF PROPOSITION A.4 PARTS 1 AND 2. Positivity:

The positivity property holds, as Definition 3.5 defines $\mathcal{U}_R(\mathcal{F}, D) = |D \div D_R|$ with an absolute value.

Monotonicity:

For monotonicity, assume that $\mathcal{F} \subseteq \hat{\mathcal{F}}$ and that D_R and $\hat{D}_R$ are the minimum repaired databases w.r.t. D and $\mathcal{F}, \hat{\mathcal{F}}$ respectively.

By definition $\hat{D}_R$ satisfies $\hat{\mathcal{F}}$ and thus in particular satisfies $\mathcal{F}$. Hence, $|D \div \hat{D}_R|$ can only be equal or larger than $|D \div D_R|$, since by our definition D_R is the database that **minimizes** the symmetric difference with D and satisfies $\mathcal{F}$. Therefore, $\mathcal{U}_R(\mathcal{F}, D) = |D \div D_R| \leq |D \div \hat{D}_R| = \mathcal{U}_R(\hat{\mathcal{F}}, D)$. $\square$

We will first state and prove the range and the sensitivity bounds of $\mathcal{U}_R$ for a single fairness criterion, and then we will use them to prove the range and the sensitivity bounds for a set of criteria (Parts 3 and 4 of Proposition A.4).

LEMMA A.5 ($\mathcal{U}_R$ IS SUITABLE FOR DP FOR A SINGLE FAIRNESS CRITERION). *Given a database D of size n , the following holds:*

- (1) The range of $\mathcal{U}_R(F, D)$ is $[0, n]$ for a single criterion F .
- (2) The sensitivity of $\mathcal{U}_R$ is at most 1 for a single criterion F .

PROOF. Range:

It is immediate that $\text{Range}(\mathcal{U}_R(F, D)) = [0, n]$ because $\mathcal{U}_R(F, D) \geq 0$ from the Positivity property, and $\mathcal{U}_R(F, D) \leq n$ because in the most extreme case, deleting all tuples in D will create the empty database that satisfies F .

Sensitivity:

Given F and $D \sim D'$ where without loss of generality $D' = D \setminus \{t'\} \cup \{t''\}$ for some tuples t' and t'' , we have $\mathcal{U}_R(F, D) = |D \div D_R|$, where D_R is the database that satisfies F with the minimum number of deletions and additions.

Since D' differs from D by replacing a single tuple t' with t'' , we can obtain the repair process for D' from the repair process for D as follows:

- If t'' interferes with F , replace t'' with t' to obtain D , and then do the repair for D . In this case $\mathcal{U}_R(F, D') = \mathcal{U}_R(F, D) + 1$.
- Otherwise, t'' does not interfere with F , and so in this case $\mathcal{U}_R(F, D') = \mathcal{U}_R(F, D)$.

Therefore, $\Delta_{\mathcal{U}_R(F, D)} = \max_{D' \sim D} |\mathcal{U}_R(F, D') - \mathcal{U}_R(F, D)| \leq 1$, and the sensitivity of $\mathcal{U}_R$ is at most 1 for a single criterion F . $\square$

PROOF OF PROPOSITION A.4 PARTS 3 AND 4. Range:

Again, $\text{Range}(\mathcal{U}_R(\mathcal{F}, D)) = [0, n]$ because $\mathcal{U}_R(\mathcal{F}, D) \geq 0$ from the Positivity property, and $\mathcal{U}_R(\mathcal{F}, D) \leq n$ because in the most extreme case, deleting all tuples in D will create the empty database that satisfies $\mathcal{F}$.

Sensitivity:

Let there be a database D_R with the smallest number of removed and added tuples with respect to D that satisfies all the fairness criteria $\mathcal{F}$. Then

$$\mathcal{U}_R(\mathcal{F}, D') \leq |D' \div D_R| \leq |D \div D_R| + 1 = \mathcal{U}_R(\mathcal{F}, D) + 1$$

From another side, let there be a database D'_R with the smallest number of removed and added tuples with respect to D' that satisfies all the fairness criteria $\mathcal{F}$. Then

$$\mathcal{U}_R(\mathcal{F}, D) \leq |D \div D'_R| \leq |D' \div D'_R| + 1 = \mathcal{U}_R(\mathcal{F}, D') + 1$$

Therefore

$$\Delta \mathcal{U}_R(\mathcal{F}, D) = \max_{D' \sim D} |\mathcal{U}_R(\mathcal{F}, D') - \mathcal{U}_R(\mathcal{F}, D)| \leq 1$$

□

PROOF OF PROPOSITION 3.12 PARTS 1 AND 2. **Positivity:**

Recall that by Definition 3.11, given a single fairness criterion F , $\mathcal{U}_{TC}$ is an absolute value. Since its aggregate form is a sum $\mathcal{U}_{TC}$ is always non-negative and its value is 0 only if the databases satisfies all the given fairness criteria in $\mathcal{F}$.

Monotonicity:

Monotonicity follows directly from positivity. □

We will first state and prove the range and the sensitivity bounds of $\mathcal{U}_{TC}$ for a single fairness criterion, and then we will use them to prove the range and the sensitivity bounds for a set of criteria (Parts 3 and 4 of Proposition 3.12).

LEMMA A.6 ($\mathcal{U}_{TC}$ IS SUITABLE FOR DP FOR A SINGLE FAIRNESS CRITERION). *Given a database D and a parameter $k \in \mathbb{N}$, the following holds:*

- (1) *The range of $\mathcal{U}_{TC}(F, D)$ is $[0, k]$ for a single criterion F .*
- (2) *The sensitivity of $\mathcal{U}_{TC}(F, D)$ is at most $\frac{3k}{n} + 2$ for an unconditional criterion F , and at most $\max_a \frac{3k}{|\{t \in D \mid t[A]=a\}|} + 2$ for a conditional one.*

PROOF OF LEMMA A.6 PART 1. By Definition 3.11, given a single fairness criterion F , $\mathcal{U}_{TC}$ is an absolute value. Therefore, from the properties of the absolute value, $\mathcal{U}_{TC}(F, D) \geq 0$.

And since the $\mathcal{MD}$ is at most 1, then the following holds:

$$\begin{aligned} \mathcal{U}_{TC}(F, D) &= \sum_{t=(p,y,a) \in \text{top-}k} \left| Pr_D(P=t[P], Y=t[Y] \mid A=t[A]) \right. \\ &\quad \left. - Pr_D(P=t[P] \mid A=t[A]) Pr_D(Y=t[Y] \mid A=t[A]) \right| \\ &\leq \sum_{t=(p,y,a) \in \text{top-}k} 1 \\ &= k \end{aligned}$$

□

Now we will state and prove a few lemmas that we will use in a later proof of Lemma A.6 Part 2.

LEMMA A.7 (SENSITIVITY OF THE UNCONDITIONAL EMPIRICAL PROBABILITIES). *Given a database D of size n such that the schema of D is a superset of $P \cup Y$, the sensitivity of the unconditional empirical probabilities $Pr_D(P)$, $Pr_D(Y)$ in D is at most $\frac{1}{n}$.*

PROOF. We will prove this claim for the **joint** empirical probabilities. The same calculation can be done for the marginal empirical probabilities.

Let there be $D \sim D'$ neighboring databases that differ by one tuple $t' = (p_{t'}, y_{t'})$. Assume without loss of generality that $t' \in D'$ and $t' \notin D$.

For any (p, y) , it holds that

$$Pr_D(P=p, Y=y) = \frac{|\{t \in D \mid t[P]=p, t[Y]=y\}|}{n}$$

For $Pr_{D'}$ we divide into two cases. For any $(p, y) \neq (p_{t'}, y_{t'})$, it holds that

$$\begin{aligned} Pr_{D'}(P=p, Y=y) &= \frac{|\{t \in D' \mid t[P]=p, t[Y]=y\}|}{n} \\ &= \frac{|\{t \in D \mid t[P]=p, t[Y]=y\}|}{n} \\ &\leq \frac{|\{t \in D \mid t[P]=p, t[Y]=y\}| + 1}{n} \end{aligned}$$

Otherwise, for $(p, y) = (p_{t'}, y_{t'})$, and

$$\begin{aligned} Pr_{D'}(P=p_{t'}, Y=y_{t'}) &= \frac{|\{t \in D' \mid t[P]=p_{t'}, t[Y]=y_{t'}\}|}{n} \\ &= \frac{|\{t \in D \mid t[P]=p_{t'}, t[Y]=y_{t'}\}| + 1}{n} \end{aligned}$$

Therefore, for any (p, y) , we obtain that

$$|Pr_{D'}(P=p, Y=y) - Pr_D(P=p, Y=y)| \quad (20)$$

$$\leq \left| \frac{|\{t \in D \mid t[P]=p, t[Y]=y\}| + 1}{n} - \frac{|\{t \in D \mid t[P]=p, t[Y]=y\}|}{n} \right| \quad (21)$$

$$= \left| \frac{n - |\{t \in D \mid t[P]=p, t[Y]=y\}| + 1}{n^2} \right| \quad (22)$$

$$\leq \frac{n - |\{t \in D \mid t[P]=p, t[Y]=y\}| + 1}{n^2} \quad (23)$$

$$\leq \frac{1}{n} \quad (24)$$

where the transition from (22) to (23) is due to the fact that $n \geq |\{t \in D \mid t[P]=p, t[Y]=y\}|$.

And so

$$\Delta Pr_D \leq \frac{1}{n}$$

□

Now we will state and prove a few lemmas that we will use in a later proof of Lemma A.6 Part 2.

LEMMA A.8 (SENSITIVITY OF THE CONDITIONAL EMPIRICAL PROBABILITIES). *Given a database D of size n such that the schema of D is a superset of $P \cup Y$, the sensitivity of the conditional empirical probabilities $Pr_D(P, Y \mid A)$, $Pr_D(P \mid A)$, $Pr_D(Y \mid A)$ in D is at most $\max_a \frac{1}{|\{t \in D \mid t[A]=a\}|}$.*

PROOF. We will prove this claim for the **joint** empirical probabilities. The same calculation can be done for the marginal empirical probabilities.

Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of t'' and one less occurrence of t' than D .

There are three cases. For any (p, y, a) such that $a \notin \{a_{t'}, a_{t''}\}$, it holds that

$$Pr_{D'}(P=p, Y=y \mid A=a) = \frac{|\{t \in D' \mid t[P]=p, t[Y]=y, t[A]=a\}|}{|\{t \in D' \mid t[A]=a\}|} \quad (25)$$

$$= \frac{|\{t \in D \mid t[P]=p, t[Y]=y, t[A]=a\}|}{|\{t \in D \mid t[A]=a\}|} = Pr_D(P=p, Y=y \mid A=a) \quad (26)$$

For any (p, y, a) such that $a = a_{t'}$, this bucket of tuples loses one tuple. Therefore

$$Pr_{D'}(P=p, Y=y \mid A=a) = \frac{|\{t \in D \mid t[P]=p, t[Y]=y, t[A]=a_{t'}\}| - 1}{|\{t \in D \mid t[A]=a_{t'}\}| - 1}$$

Similarly, for any (p, y, a) such that $a = a_{t''}$, it holds that

$$Pr_{D'}(P=p, Y=y \mid A=a) = \frac{|\{t \in D \mid t[P]=p, t[Y]=y, t[A]=a_{t''}\}| + 1}{|\{t \in D \mid t[A]=a_{t''}\}| + 1}$$

And so, for any (p, y, a) , we obtain that

$$|Pr_{D'}(P=p, Y=y | A=a) - Pr_D(P=p, Y=y | A=a)| \quad (27)$$

$$\leq \begin{cases} \frac{1}{|\{t \in D \mid t[A] = a_{t'}\}| - 1}, & \text{if } a = a_{t'} \text{ (the bucket that loses a tuple),} \\ \frac{1}{|\{t \in D \mid t[A] = a_{t''}\}| + 1}, & \text{if } a = a_{t''} \text{ (the bucket that gains a tuple),} \\ 0, & \text{otherwise.} \end{cases} \quad (28)$$

Therefore

$$\Delta Pr_D \leq \max_a \frac{1}{|\{t \in D \mid t[A] = a\}| - 1} < \max_a \frac{1}{|\{t \in D \mid t[A] = a\}|}.$$

□

LEMMA A.9. *Given a database D of size n such that the schema of D is a superset of $P \cup Y \cup A$, the sensitivity of the marginal difference is at most $\frac{3}{n}$ in the unconditional case and $\max_a \frac{4}{|\{t \in D \mid t[A] = a\}|}$ in the conditional case.*

PROOF. Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of t'' and one less occurrence of t' than D .

Unconditional case:

For any (p, y) , it holds that

$$|\mathcal{MD}_{D'}(P \perp Y, (p, y)) - \mathcal{MD}_D(P \perp Y, (p, y))| \quad (29)$$

$$= |(Pr_{D'}(P=p, Y=y) - Pr_{D'}(P=p) Pr_{D'}(Y=y)) - (Pr_D(P=p, Y=y) - Pr_D(P=p) Pr_D(Y=y))| \quad (30)$$

$$= |(Pr_{D'}(P=p, Y=y) - Pr_D(P=p, Y=y)) - (Pr_{D'}(P=p) Pr_{D'}(Y=y) - Pr_D(P=p) Pr_D(Y=y))| \quad (31)$$

$$= |[Pr_{D'}(P=p, Y=y) - Pr_D(P=p, Y=y)] - [Pr_{D'}(P=p) (Pr_{D'}(Y=y) - Pr_D(Y=y)) + Pr_D(Y=y) (Pr_{D'}(P=p) - Pr_D(P=p))]| \quad (32)$$

$$\leq |[Pr_{D'}(P=p, Y=y) - Pr_D(P=p, Y=y)] + [Pr_{D'}(P=p) (Pr_{D'}(Y=y) - Pr_D(Y=y)) + Pr_D(Y=y) (Pr_{D'}(P=p) - Pr_D(P=p))]| \quad (33)$$

$$\leq |Pr_{D'}(P=p, Y=y) - Pr_D(P=p, Y=y)| + |Pr_{D'}(P=p) (Pr_{D'}(Y=y) - Pr_D(Y=y))| + |Pr_D(Y=y) (Pr_{D'}(P=p) - Pr_D(P=p))| \quad (34)$$

$$\leq \frac{1}{n} + |Pr_{D'}(P=p)| \cdot \frac{1}{n} + |Pr_D(Y=y)| \cdot \frac{1}{n} \quad (35)$$

$$\leq \frac{3}{n} \quad (36)$$

where the transitions are due to:

- (31) $\rightarrow$ (32): $ab - cd = a(b - d) + d(a - c)$.
- (33) $\rightarrow$ (34): triangle inequality.
- (34) $\rightarrow$ (35): Lemma A.7.
- (35) $\rightarrow$ (36): $Pr_{D'}(P=p) \leq 1, Pr_D(Y=y) \leq 1$.

And so

$$\Delta \mathcal{MD}_D(F, t) \leq \frac{3}{n}$$

Conditional case:

By Lemma A.8, each conditional probability term can change by at most $\max_a \frac{1}{|\{t \in D \mid t[A] = a\}|}$. Therefore, similarly to the unconditional case, for any (p, y, a) by the definition of conditional $\mathcal{MD}$ it

holds that

$$\begin{aligned} & |\mathcal{MD}_{D'}(P \perp Y | A, t) - \mathcal{MD}_D(P \perp Y | A, t)| \\ &= |Pr_{D'}(A=a)(Pr_{D'}(P=p, Y=y | A=a) - Pr_{D'}(P=p | A=a) Pr_{D'}(Y=y | A=a)) \\ &\quad - Pr_D(A=a)(Pr_D(P=p, Y=y | A=a) - Pr_D(P=p | A=a) Pr_D(Y=y | A=a))| \\ &\leq |a'b' - ab| \leq |a' - a| + |b' - b| |Pr_{D'}(A=a) - Pr_D(A=a)| \\ &\quad + |(Pr_{D'}(P=p, Y=y | A=a) - Pr_{D'}(P=p | A=a) Pr_{D'}(Y=y | A=a)) \\ &\quad - (Pr_D(P=p, Y=y | A=a) - Pr_D(P=p | A=a) Pr_D(Y=y | A=a))| \\ &\leq 4 \cdot \max_a \frac{1}{|\{t \in D \mid t[A] = a\}|} \\ &\leq \max_a \frac{4}{|\{t \in D \mid t[A] = a\}|} \end{aligned}$$

□

PROOF OF LEMMA A.6 PART 2. **Unconditional case:**

Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of $t'' = (p_{t''}, y_{t''})$ and one less occurrence of $t' = (p_{t'}, y_{t'})$ than D .

We can notice that the absolute difference between the contribution measure values between D and D' is the biggest when the difference between the top- k lists based on D and D' are the two respective tuples t' and t'' . Therefore, the sensitivity is bounded by the sum of the differences $|\mathcal{MD}_{D'}(F, t) - \mathcal{MD}_D(F, t)|$ for every t in the intersection of the top- k lists, plus 2 for t' and t'' . And so

$$|\mathcal{U}_{TC}(F, D') - \mathcal{U}_{TC}(F, D)| \quad (37)$$

$$= \left| \sum_{t \in \text{top-}k'} \mathcal{MD}_{D'}(F, t) - \sum_{t \in \text{top-}k} \mathcal{MD}_D(F, t) \right| \quad (38)$$

$$\leq \left| \sum_{t \in \text{top-}k' \cap \text{top-}k} \mathcal{MD}_{D'}(F, t) - \sum_{t \in \text{top-}k' \cap \text{top-}k} \mathcal{MD}_D(F, t) \right| + 2 \quad (39)$$

$$= \left| \sum_{t \in \text{top-}k' \cap \text{top-}k} (\mathcal{MD}_{D'}(F, t) - \mathcal{MD}_D(F, t)) \right| + 2 \quad (40)$$

$$\leq \sum_{t \in \text{top-}k' \cap \text{top-}k} |\mathcal{MD}_{D'}(F, t) - \mathcal{MD}_D(F, t)| + 2 \quad (41)$$

$$\leq k \cdot |\mathcal{MD}_{D'}(F, t) - \mathcal{MD}_D(F, t)| + 2 \quad (42)$$

$$\leq k \cdot \Delta \mathcal{MD}_D(F, t) + 2 \quad (43)$$

$$\leq k \cdot \frac{3}{n} + 2 \quad (44)$$

$$= \frac{3k}{n} + 2 \quad (45)$$

where the transitions are due to:

- (38) $\rightarrow$ (39): explanation earlier.
- (40) $\rightarrow$ (41): triangle inequality.
- (41) $\rightarrow$ (42): $|\text{top-}k' \cap \text{top-}k| \leq k$.
- (43) $\rightarrow$ (44): Lemma A.9.

Conditional case:

Similarly, assume that D' contains one more occurrence of $t'' = (p_{t''}, y_{t''}, a_{t''})$ and one less occurrence of $t' = (p_{t'}, y_{t'}, a_{t'})$ than D .

Some of the transitions are identical as in the unconditional case and, therefore, are skipped.

Denote:

$$\begin{aligned} n'_{a_{t'}} &= |t \in D' \mid t[A] = a_{t'}|, & n'_{a_{t''}} &= |t \in D' \mid t[A] = a_{t''}| \\ n_{a_{t'}} &= |t \in D \mid t[A] = a_{t'}|, & n_{a_{t''}} &= |t \in D \mid t[A] = a_{t''}| \end{aligned}$$

$$\begin{aligned}
 & |\mathcal{U}_{TC}(F, D') - \mathcal{U}_{TC}(F, D)| \\
 &= \left| \sum_{t \in \text{top-}k'} \mathcal{MD}_{D'}(F, t) - \sum_{t \in \text{top-}k} \mathcal{MD}_D(F, t) \right| \\
 &\leq \sum_{t \in \text{top-}k' \cap \text{top-}k} |\mathcal{MD}_{D'}(F, t) - \mathcal{MD}_D(F, t)| + 2 \\
 &\leq k \cdot |\mathcal{MD}_{D'}(F, t) - \mathcal{MD}_D(F, t)| + 2 \\
 &\leq k \cdot \Delta_{\mathcal{MD}_D}(F, t) + 2 \\
 &\leq \max_a \frac{4k}{|\{t \in D \mid t[A] = a\}|} + 2
 \end{aligned}$$

where $\Delta_{\mathcal{MD}_D}(F, t) \leq \max_a \frac{4}{|\{t \in D \mid t[A] = a\}|}$ by Lemma A.9. $\square$

The proof of Proposition 3.12 Parts 3 and 4 is similar to the respective parts of the proof of Proposition 3.4.

A.2 Proofs for Section 4

PROOF OF LEMMA 3.8. Let D_{α_i} be the set of tuples t such that $x_t = \text{True}$ under assignment α_i .

Recall that the repair cost is defined as the symmetric difference between D and D_{α_i}

$$\mathcal{U}_R(\varphi(D, D_{S_j}), \alpha_i) = |D \setminus D_{\alpha_i}| + |D_{\alpha_i} \setminus D|$$

This expression counts deletions (tuples in D but not in D_{α_i}) and insertions (tuples in D_{α_i} but not in D). Now, observe that:

- Each satisfied soft clause of the form x_t (with $t \in D$) means that t is retained in D_{α_i} (i.e., not deleted).
- Each satisfied soft clause of the form $\neg x_t$ (with $t \in D_{S_j} \setminus D$) means that t is not inserted into D_{α_i} .

Thus, satisfying more soft clauses corresponds to performing fewer changes to the original database. Specifically, more original tuples are kept, and fewer new tuples are added. Hence, the symmetric difference $|D \setminus D_{\alpha_i}| + |D_{\alpha_i} \setminus D|$ is smaller.

Therefore, if α_1 satisfies more soft clauses than α_2 , it must result in a smaller repair cost by the definition of $\mathcal{U}_R$. That is

$$\Delta(\varphi(D, D_{S_j}), \alpha_1) < \Delta(\varphi(D, D_{S_j}), \alpha_2)$$

$\square$

PROOF OF ???. By Definition 3.6, each tuple $t \in D$ contributes a soft clause x_t , and each tuple $t \in D_{S_j} \setminus D$ contributes a soft clause $\neg x_t$.

For any feasible assignment α , let $D_R = \{t \in D_{S_j} \mid \alpha(x_t) = \text{True}\}$. Then, since soft clauses are a multiset, a soft clause is not satisfied iff the respective tuple appears in D and doesn't appear in D_R . That is, the number of satisfied soft clauses is

$$\#\{\text{soft clauses satisfied by } \alpha\} = |S(D_{S_j})| - \Delta(\varphi(D, D_{S_j}), \alpha)$$

Therefore, minimizing $\Delta(\varphi(D, D_{S_j}), \alpha)$ is equivalent to maximizing the number of satisfied soft clauses. Thus, the two definitions are also equivalent. $\square$

Following are definitions and propositions with proofs for the full $\mathcal{U}_R^{S\mathcal{AT}}$ sensitivity analysis.

DEFINITION A.10 (ASSIGNMENT EXTENSION). Given a database D and a CNF formula $\varphi(D, D_{S_j}) = \mathcal{H}(D_{S_j}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{S_j} \setminus D} (\neg x_t)$, an assignment α for $\varphi(D, D_{S_j})$, and a CNF formula $\varphi' = \mathcal{H}(D_{\text{cross}}) \wedge$

$\bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{\text{cross}} \setminus D} (\neg x_t)$ such that $D_{S_j} \subseteq D_{\text{cross}}$, an extension of the assignment α for $\varphi(D, D_{\text{cross}})$ is defined as follows:

$$\alpha'(x_t) = \begin{cases} \alpha(x_t) & \text{if } t \in D_{S_j} \\ \text{False} & \text{if } t \in D_{\text{cross}} \setminus D_{S_j} \end{cases}$$

With this definition, we show that a minimum repair of the join and cross-product databases by their CNF formulae have identical size. We then prove that extending a CNF formula for the join database to a CNF formula for the cross-product database preserves hard clauses satisfaction.

LEMMA A.11. Given a database D and a fairness criterion, let D_{S_j} and D_{cross} be the self-join and cross-product databases. Let $\varphi(D, D_{S_j})$ and $\varphi(D, D_{\text{cross}})$ be two CNF formulae defined according to Definition 3.6. Let α be an optimal assignment for $\varphi(D, D_{S_j})$, and let $\hat{\alpha}$ be its extension to the assignment for $\varphi(D, D_{\text{cross}})$ according to Definition A.13. It holds that $\mathcal{U}_R^{S\mathcal{AT}}(\varphi(D, D_{S_j})) = \mathcal{U}_R^{S\mathcal{AT}}(\varphi(D, D_{\text{cross}}))$.

PROOF. Let α be an optimal assignment for $\varphi(D, D_{S_j})$, and let $\hat{\alpha}$ be its extension to $\varphi(D, \hat{D})$, as defined in Definition A.13.

Since α is an optimal assignment, it satisfies all hard clauses in $\varphi(D, D_{S_j})$. Therefore, by Lemma A.12, $\hat{\alpha}$ satisfies all hard clauses in $\varphi(D, \hat{D})$. Additionally, by Lemma A.14, the number of satisfied soft clauses in $\varphi(D, \hat{D})$ under $\hat{\alpha}$ is at least that in $\varphi(D, D_{S_j})$ under α . Hence, $\hat{\alpha}$ is a feasible and possibly optimal assignment for $\varphi(D, \hat{D})$.

Therefore, by Lemma 3.8 it holds that

$$\Delta(\varphi(D, \hat{D}), \hat{\alpha}) \leq \Delta(\varphi(D, D_{S_j}), \alpha)$$

Now let $\hat{\alpha}^*$ be an optimal assignment for $\varphi(D, \hat{D})$, and let α^* be its restriction to D_{S_j} defined as follows

$$\alpha^*(x_t) = \hat{\alpha}^*(x_t) \text{ for } t \in D_{S_j}$$

Since $D_{S_j} \subseteq \hat{D}$, then $\mathcal{H}(D_{S_j}) \subseteq \mathcal{H}(\hat{D})$. And because hard clause satisfaction is preserved under restriction (Lemma A.12), it follows that α^* is a feasible assignment for $\varphi(D, D_{S_j})$.

Moreover, the reverse direction of Lemma A.14 implies that the number of soft clauses satisfied under α^* is at least that in $\varphi(D, D_{S_j})$ under any other assignment. This means that α^* is an optimal assignment α for $\varphi(D, D_{S_j})$, and $\hat{\alpha}$ is an optimal assignment $\hat{\alpha}^*$ for $\varphi(D, \hat{D})$ by Definition A.13.

Therefore, by Lemma 3.8 it holds that

$$\Delta(\varphi(D, D_{S_j}), \alpha) = \Delta(\varphi(D, D_{S_j}), \alpha^*) \leq \Delta(\varphi(D, \hat{D}), \hat{\alpha}) = \Delta(\varphi(D, \hat{D}), \hat{\alpha}^*)$$

So from Definition 3.9 and from both directions we proved, it holds that

$$\begin{aligned}
 \mathcal{U}_R^{S\mathcal{AT}}(F, D) &= \Delta(\varphi(D, D_{S_j}), \alpha) \\
 &= \Delta(\varphi(D, \hat{D}), \hat{\alpha}) = \min_{\alpha \models \mathcal{H}(\hat{D})} \Delta(\varphi(D, \hat{D}), \alpha)
 \end{aligned}$$

$\square$

We will now prove that extending a CNF formula for the join database to a CNF formula for the cross-product database preserves hard clauses satisfaction.

LEMMA A.12. Given a database D such that the schema of D is a superset of $P \cup Y \cup A$, and a fairness criterion of the form $F = P \sqcup Y \mid A$, let D_{S_j} be the join database and D_{cross} be cross-product database, and let $\varphi(D, D_{S_j})$ and $\varphi(D, D_{\text{cross}})$ be the corresponding two CNF formulae from Definition 3.6. Let α be a feasible assignment for $\varphi(D, D_{S_j})$ and let $\hat{\alpha}$ be its extension to the assignment for $\varphi(D, D_{\text{cross}})$ according to Definition A.13. Then, $\mathcal{H}(D_{S_j})$ in $\varphi(D, D_{S_j})$ are satisfied by α iff $\mathcal{H}(D_{S_j})$ in $\varphi(D, D_{\text{cross}})$ are satisfied by $\hat{\alpha}$.

PROOF. ($\Rightarrow$) Suppose that $\mathcal{H}(D_{sj})$ in $\varphi(D, D_{sj})$ are satisfied under α . Since $\hat{D}$ is a superset of D_{sj} , then each clause in $\mathcal{H}(D_{sj})$ appears in $\mathcal{H}(\hat{D}_{sj})$, and they differ only in additional hard clauses added because of the tuples in $\hat{D} \setminus D_{sj}$. Since $\hat{\alpha}$ agrees with α on all $t \in D_{sj}$, the same clauses also evaluate to True under $\hat{\alpha}$.

Now consider the additional clauses introduced due to tuples in $\hat{D} \setminus D_{sj}$. These clauses are of the form $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$, where at least one of the tuples t_1, t_2, t_3 is in $\hat{D} \setminus D_{sj}$. Since $\hat{\alpha}$ assigns False to all x_t so that $t \in \hat{D} \setminus D_{sj}$, the clause $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$ will evaluate to True as follows

- If either x_{t_1} or x_{t_2} is False under $\hat{\alpha}$, their negation is True, so the whole clause is satisfied by $\hat{\alpha}$.
- If both x_{t_1} and x_{t_2} are True under $\hat{\alpha}$, this is possible only if both t_1 and t_2 are in D_{sj} .

According to the construction of the CNF formula (see Definition 3.6), a clause of the form

$$(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$$

is included in the hard clauses, where $t_1 = (P_1, Y_1, A) \in D_{sj}$, $t_2 = (P_2, Y_2, A) \in D_{sj}$, and $t_3 = (P_1, Y_2, A)$ is the tuple resulting from the join of t_1 and t_2 on the shared attribute A .

Since both t_1 and t_2 belong to D_{sj} , and the join $D_{sj}(P_1, Y_2, A) := D(P_1, Y_1, A) \bowtie D(P_2, Y_2, A)$ preserves A , the resulting tuple $t_3 = (P_1, Y_2, A)$ must also belong to D_{sj} . Therefore, the clause $(\neg x_{t_1} \vee \neg x_{t_2} \vee x_{t_3})$ belongs to $\mathcal{H}(D_{sj})$, and since $\hat{\alpha}$ agrees with α on all tuples in D_{sj} , it satisfies this clause as well. Then the full clause is in $\mathcal{H}(D_{sj})$ and should be satisfied by $\hat{\alpha}$ because it agrees with α on all $t \in D_{sj}$.

Therefore, in any case $\mathcal{H}(\hat{D})$ in $\varphi(D, \hat{D})$ are satisfied under $\hat{\alpha}$.

($\Leftarrow$) Suppose $\mathcal{H}(\hat{D})$ in $\varphi(D, \hat{D})$ are satisfied under $\hat{\alpha}$. In particular, all clauses in $\mathcal{H}(D_{sj})$, which are a subset of $\mathcal{H}(\hat{D})$, are satisfied under $\hat{\alpha}$. Since $\hat{\alpha}$ agrees with α on all x_t for $t \in D_{sj}$, then $\mathcal{H}(D_{sj})$ in $\varphi(D, D_{sj})$ are satisfied also under α .

From the two directions, it holds that $\mathcal{H}(D_{sj})$ in $\varphi(D, D_{sj})$ are satisfied under α iff $\mathcal{H}(\hat{D})$ in $\varphi(D, \hat{D})$ are satisfied under $\hat{\alpha}$. $\square$

Finally, we show that extending a CNF formula for the self-join database to a CNF formula for the cross-product database cannot decrease the number of satisfied soft clauses.

DEFINITION A.13 (ASSIGNMENT EXTENSION). *Given a database D and a CNF formula $\varphi(D, D_{sj}) = \mathcal{H}(D_{sj}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj}-D} (\neg x_t)$, an assignment α for $\varphi(D, D_{sj})$, and a CNF formula $\varphi' = \mathcal{H}(D_{cross}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{cross}-D} (\neg x_t)$ such that $D_{sj} \subseteq D_{cross}$, an extension of the assignment α for $\varphi(D, D_{cross})$ is defined as follows:*

$$\alpha'(x_t) = \begin{cases} \alpha(x_t) & \text{if } t \in D_{sj} \\ \text{False} & \text{if } t \in D_{cross} \setminus D_{sj} \end{cases}$$

LEMMA A.14. *Given a database D and a fairness criterion, let D_{sj} and D_{cross} be the join and cross-product databases. Let $\varphi(D, D_{sj})$ and $\varphi(D, D_{cross})$ be the two corresponding CNF formulae according to Definition 3.6. Let α be a feasible assignment for $\varphi(D, D_{sj})$ and let $\hat{\alpha}$ be its extension to the assignment for $\varphi(D, D_{cross})$. Then the number of satisfied soft clauses in $\varphi(D, D_{sj})$ by α is lower than or equal to the number of satisfied soft clauses in $\varphi(D, D_{cross})$ under $\hat{\alpha}$.*

PROOF. Let D be a dataset. According to Definition 3.6:

$$\begin{aligned} \varphi(D, D_{sj}) &= \mathcal{H}(D_{sj}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in D_{sj} \setminus D} \neg x_t \\ \varphi(D, \hat{D}) &= \mathcal{H}(\hat{D}) \wedge \bigwedge_{t \in D} x_t \wedge \bigwedge_{t \in \hat{D} \setminus D} \neg x_t \end{aligned}$$

Since $D_{sj} \subseteq \hat{D}$, then $\varphi(D, D_{sj})$ and $\varphi(D, \hat{D})$ differ only in soft clauses for the tuples in $\hat{D} \setminus D_{sj}$. In particular, the only difference is that $\varphi(D, \hat{D})$ contains additional soft clauses of the form $\neg x_t$ for $t \in \hat{D} \setminus D_{sj}$.

Let α be an assignment for $\varphi(D, D_{sj})$. According to Definition A.13, it holds that

- All soft clauses in $\varphi(D, D_{sj})$ are also present in $\varphi(D, \hat{D})$ and are satisfied under $\hat{\alpha}$ exactly as in α .
- The additional soft clauses in $\varphi(D, \hat{D})$ are of the form $\neg x_t$ for $t \in \hat{D} \setminus D_{sj}$, and each such clause is satisfied in $\hat{\alpha}$ because $\hat{\alpha}(x_t) = \text{False}$.

Hence, $\hat{\alpha}$ satisfies all the soft clauses that α satisfies, plus potentially some new ones. Therefore, the number of satisfied soft clauses in $\varphi(D, D_{sj})$ under α is less or equal to the number of satisfied soft clauses in $\varphi(D, \hat{D})$ under $\hat{\alpha}$. $\square$

As a precursor to the sensitivity analysis, we show that the difference in repair between neighboring databases is bounded by 2.

LEMMA A.15. *Let $D \sim D'$, let D_{cross} and D'_{cross} be their corresponding cross-product databases, and let $F = P \sqcup Y \mid A$ be a fairness criterion. Let $\varphi(D, D_{cross})$ and $\varphi(D', D'_{cross})$ be CNF formulae defined according to Definition 3.6, and let $\hat{\alpha}$ be a feasible assignment for $\varphi(D, D_{cross})$. Finally, let $\Delta(\varphi(D, D_{cross}), \hat{\alpha})$ and $\Delta(\varphi(D', D'_{cross}), \hat{\alpha})$ be defined according to Definition 3.7. It holds that:*

$$|\Delta(\varphi(D, D_{cross}), \hat{\alpha}) - \Delta(\varphi(D', D'_{cross}), \hat{\alpha})| \leq 2$$

PROOF. Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of t'' and one less occurrence of t' than D .

It follows that the only difference between $\varphi(D, \hat{D})$ and $\varphi(D', \hat{D}')$ lies at most in two soft clauses:

- If $t' \notin D'$, then $\varphi(D', \hat{D}')$ contains the soft clause $\neg x_{t'}$, while $\varphi(D, \hat{D})$ contains the soft clause $x_{t'}$.
- If $t'' \notin D$, then $\varphi(D', \hat{D}')$ contains the soft clause $x_{t''}$, while $\varphi(D, \hat{D})$ contains the soft clause $\neg x_{t''}$.

The set of hard clauses remains the same, as they are defined over $\hat{D}$, which depends only on the attribute domains. And since there is no difference in the hard clauses between D and D' , then $\hat{\alpha}$ is also a feasible assignment for $\varphi(D', \hat{D}')$.

Now, observe that under $\hat{\alpha}$

- If $\hat{\alpha}(x_{t'}) = \text{True}$, then $\varphi(D', \hat{D}')$ gains one satisfied soft clause ($x_{t'}$), and $\varphi(D, \hat{D})$ loses one satisfied soft clause ($\neg x_{t'}$ violated).
- If $\hat{\alpha}(x_{t'}) = \text{False}$, then $\varphi(D', \hat{D}')$ loses one soft clause and $\varphi(D, \hat{D})$ gains one.

Thus, the number of satisfied soft clauses under $\hat{\alpha}$ differs by at most 2 between the two formulas. As a result, the sets of tuples $D_{\hat{\alpha}}$ (tuples with $x_t = \text{True}$) used to compute the repair values may differ by at most one tuple, and the database D differs from D' by one tuple as well. Therefore, the symmetric difference

$$|D \setminus D_{\hat{\alpha}}| + |D_{\hat{\alpha}} \setminus D| \quad \text{vs} \quad |D' \setminus D_{\hat{\alpha}}| + |D_{\hat{\alpha}} \setminus D'|$$

can differ by at most 2.

Hence

$$|\Delta(\varphi(D, \hat{\alpha}), \hat{\alpha}) - \Delta(\varphi(D', \hat{\alpha}), \hat{\alpha})| \leq 2$$

□

We are now ready to combine the lemmas into our result that bounds the sensitivity of $\mathcal{U}_R^{\text{SAT}}$.

PROPOSITION A.16 (SENSITIVITY OF $\mathcal{U}_R^{\text{SAT}}$). *For a database D and a fairness criterion F , the sensitivity of $\mathcal{U}_R^{\text{SAT}}(F, D)$ is at most 2.*

PROOF OF PROPOSITION A.16. Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Without loss of generality, assume that D' contains one more occurrence of t'' and one less occurrence of t' than D .

Given a fairness criterion of the form $F = P \perp Y \mid A$, let D_{sj} and $\hat{D}$ be the join and cross-product databases, and similarly for D' .

Since $\hat{D} = \hat{D}'$, (the cross-product database depends only on domains), the set of hard clauses is identical in both formulas $\varphi(D, \hat{D})$ and $\varphi(D', \hat{D}')$. We can also notice that the only difference between these formulas lies in at most two soft clauses, as formulated in the proof of Lemma A.15.

By Lemma A.11,

$$\mathcal{U}_R^{\text{SAT}}(F, D) = \min_{\alpha \models H(\hat{D})} \Delta(\varphi(D, \hat{D}), \alpha), \quad \mathcal{U}_R^{\text{SAT}}(F, D') = \min_{\alpha \models H(\hat{D})} \Delta(\varphi(D', \hat{D}), \alpha)$$

Let

$$\alpha_D \in \arg \min_{\alpha \models H(\hat{D})} \Delta(\varphi(D, \hat{D}), \alpha)$$

and

$$\alpha_{D'} \in \arg \min_{\alpha \models H(\hat{D})} \Delta(\varphi(D', \hat{D}), \alpha)$$

be optimal assignments.

By Lemma A.15, for every feasible α , it holds that

$$|\Delta(\varphi(D, \hat{D}), \alpha) - \Delta(\varphi(D', \hat{D}), \alpha)| \leq 2$$

Applying this bound with $\alpha = \alpha_D$ and using optimality of the assignments, we get

$$\begin{aligned} \mathcal{U}_R^{\text{SAT}}(F, D) &= \Delta(\varphi(D, \hat{D}), \alpha_D) \\ &\leq \Delta(\varphi(D', \hat{D}), \alpha_D) + 2 \\ &\leq \Delta(\varphi(D', \hat{D}), \alpha_{D'}) + 2 \\ &= \mathcal{U}_R^{\text{SAT}}(F, D') + 2 \end{aligned}$$

By symmetry (if we swap D and D'), we also have $\mathcal{U}_R^{\text{SAT}}(F, D') \leq \mathcal{U}_R^{\text{SAT}}(F, D) + 2$. Therefore, we get

$$\Delta_{\mathcal{U}_R^{\text{SAT}}(F, D)} = \max_{D' \sim D} |\mathcal{U}_R^{\text{SAT}}(F, D) - \mathcal{U}_R^{\text{SAT}}(F, D')| \leq 2$$

□

PROOF OF PROPOSITION 3.10 PART 4. According to Proposition A.16, for a single fairness criterion it holds that the difference between the soft clauses in the CNFs for two neighboring databases is at most one, and also that the difference in the cost of repair is at most 2. So for a set of fairness criteria $\mathcal{F}$ it holds that the difference between the soft clauses in the CNFs is at most $|\mathcal{F}|$.

By repeating the proof of Proposition A.16, we can obtain that the difference in the cost of repair between two neighboring databases for a set of fairness criteria $\mathcal{F}$ is at most $2|\mathcal{F}|$. □

A.3 Figures and Proofs for Section 5

Comparing $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ and $\mathcal{U}_{\text{TC}}$.

In Figure 9, we compared $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ and $\mathcal{U}_{\text{TC}}$ without privacy considerations. While the values of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ are smaller than the values of $\mathcal{U}_{\text{TC}}$ for all criteria, it can be noticed that $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ and $\mathcal{U}_{\text{TC}}$ exhibited exactly the same trends in values. That is, higher values of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ correspond to higher values of $\mathcal{U}_{\text{TC}}$, and vice versa. Moreover, relative differences in the values of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ are reflected by comparable relative differences in the values of $\mathcal{U}_{\text{TC}}$.

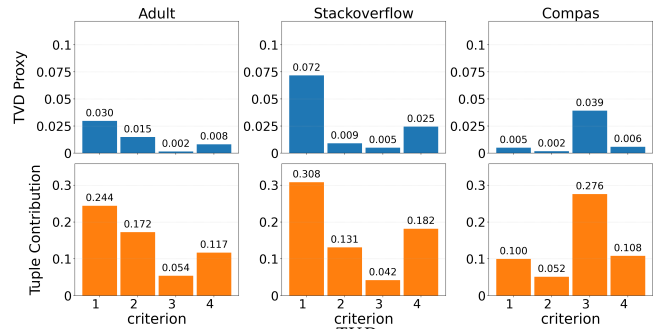


Figure 9: Comparison of $\mathcal{U}_{\text{MI}}^{\text{TVD}}$ and $\mathcal{U}_{\text{TC}}$ without privacy considerations.

Comparing $\mathcal{U}_R^{\text{SAT}}$ with and without the heuristic.

In this section we will denote $\mathcal{U}_R^{\text{SAT}}$ with the chunking heuristic as $\mathcal{U}_R^{\text{SAT}_{\text{chunk}}}$. We will prove that the sensitivity bound of $\mathcal{U}_R^{\text{SAT}_{\text{chunk}}}$ is the same as the sensitivity bound of $\mathcal{U}_R^{\text{SAT}}$.

PROPOSITION A.17 (SENSITIVITY OF $\mathcal{U}_R^{\text{SAT}_{\text{chunk}}}$). *For a database D and a fairness criterion F , the sensitivity of $\mathcal{U}_R^{\text{SAT}_{\text{chunk}}}(F, D)$ is at most 2.*

PROOF. Let there be $D \sim D'$ neighboring databases such that they differ in one tuple by replacement. Let $\text{Chunk}(\cdot)$ be the chunking procedure used by $\mathcal{U}_R^{\text{SAT}_{\text{chunk}}}$, which consists of sorting and then partitioning into consecutive blocks of fixed size, such that we assume that all duplicates of any tuple are placed into the same block, the blocks are disjoint and their union equals the full database. We write

$$\text{Chunk}(D) = \{D^{(1)}, \dots, D^{(m)}\}, \quad \text{Chunk}(D') = \{D'^{(1)}, \dots, D'^{(m)}\}$$

By the definition of $\mathcal{U}_R^{\text{SAT}_{\text{chunk}}}$, we assume that any neighbor D' of D has the same tuple ordering as D . Therefore, the replacement of a single tuple can affect the contents of at most one chunk, which is the corresponding chunk in D' . Hence, there exists an index $j \in \{1, \dots, m\}$ such that

$$D^{(i)} = D'^{(i)} \text{ for all } i \neq j, \quad D^{(j)} \sim D'^{(j)}$$

By the definition of $\mathcal{U}_R^{\text{SAT}_{\text{chunk}}}$, it holds that

$$\mathcal{U}_R^{\text{SAT}_{\text{chunk}}}(F, D) = \sum_{i=1}^m \mathcal{U}_R^{\text{SAT}}(F, D^{(i)}), \quad \mathcal{U}_R^{\text{SAT}_{\text{chunk}}}(F, D') = \sum_{i=1}^m \mathcal{U}_R^{\text{SAT}}(F, D'^{(i)})$$

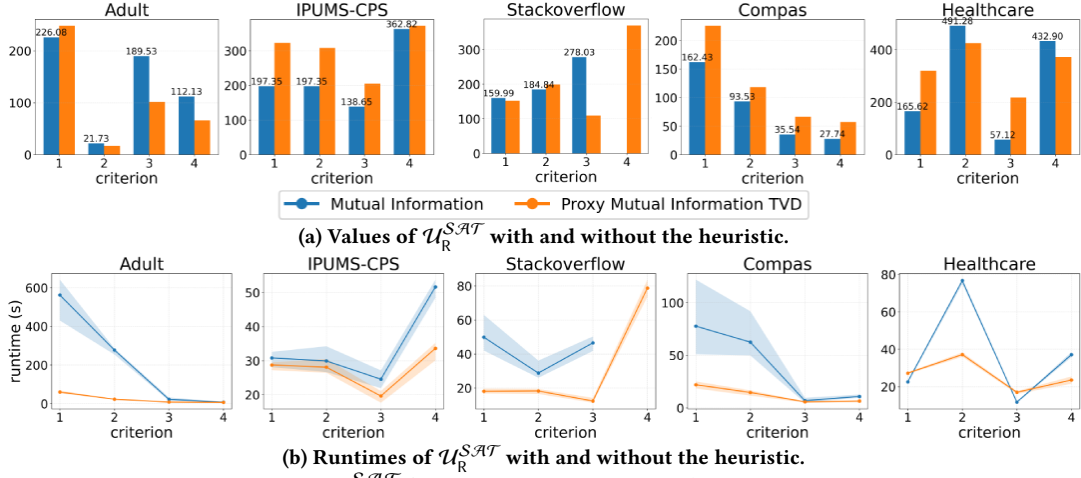


Figure 10: Comparison of $\mathcal{U}_R^{SAT}$ (computed by Algorithm 2) with and without the heuristic.

Therefore,

$$\begin{aligned} \left| \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D) - \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D') \right| &= \left| \sum_{i=1}^m \mathcal{U}_R^{SAT}(F, D^{(i)}) - \sum_{i=1}^m \mathcal{U}_R^{SAT}(F, D'^{(i)}) \right| \\ &= \left| \mathcal{U}_R^{SAT}(F, D^{(j)}) - \mathcal{U}_R^{SAT}(F, D'^{(j)}) \right| \end{aligned}$$

Since $D^{(j)} \sim D'^{(j)}$, by Proposition A.16) we get

$$\left| \mathcal{U}_R^{SAT}(F, D^{(j)}) - \mathcal{U}_R^{SAT}(F, D'^{(j)}) \right| \leq 2.$$

Combining the above yields

$$\left| \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D) - \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D') \right| \leq 2$$

Therefore, we get

$$\Delta_{\mathcal{U}_R^{SAT_{\text{chunk}}}(F, D)} = \max_{D' \sim D} \left| \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D) - \mathcal{U}_R^{SAT_{\text{chunk}}}(F, D') \right| \leq 2$$

□

In Figure 10, we compared the versions of Algorithm 2 with and without the heuristic (that is, computing $\mathcal{U}_R^{SAT_{\text{chunk}}}$ with chunks of size 100, and $\mathcal{U}_R^{SAT}$, respectively), with privacy budget $\epsilon = 1$. The algorithm for $\mathcal{U}_R^{SAT_{\text{chunk}}}$ was noticeably faster than for $\mathcal{U}_R^{SAT}$. It can also be noticed that the values of $\mathcal{U}_R^{SAT_{\text{chunk}}}$ almost exactly mirror the trend of the values of $\mathcal{U}_R^{SAT}$, except for the third criterion for Stackoverflow.